

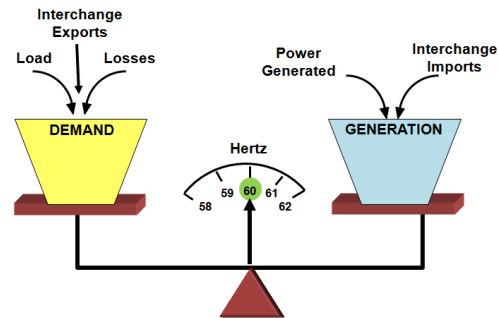


NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment

Ancillary Services

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Note: presentation is significantly updated compared with book. Where there are inconsistencies between the book and the presentation, the latter prevails

Overview

- What are "ancillary services"
- Reference model for active reserves
- Frequency-supporting ancillary services
- Simple model for pricing of active reserves
- Reserve markets in the Nordic system
- Upcoming pan-European balancing platforms

Ancillary Services - definition

IEC (International Electrotechnical Commission) – Electropedia¹ (IEV online), Section 617-03-09: Market organization and technical constraints:

- Services necessary for the operation of an electric power system provided by the system operator and/or by power system users

Note – Ancillary services may include the participation in frequency regulation, reactive power regulation, active power reservation, etc.

- *“Ancillary services are all services required by the transmission or distribution system operator to enable them to maintain the integrity and stability of the transmission or distribution system as well as the power quality.” (EURELECTRIC)*

¹ <https://www.electropedia.org/>

Ancillary Services

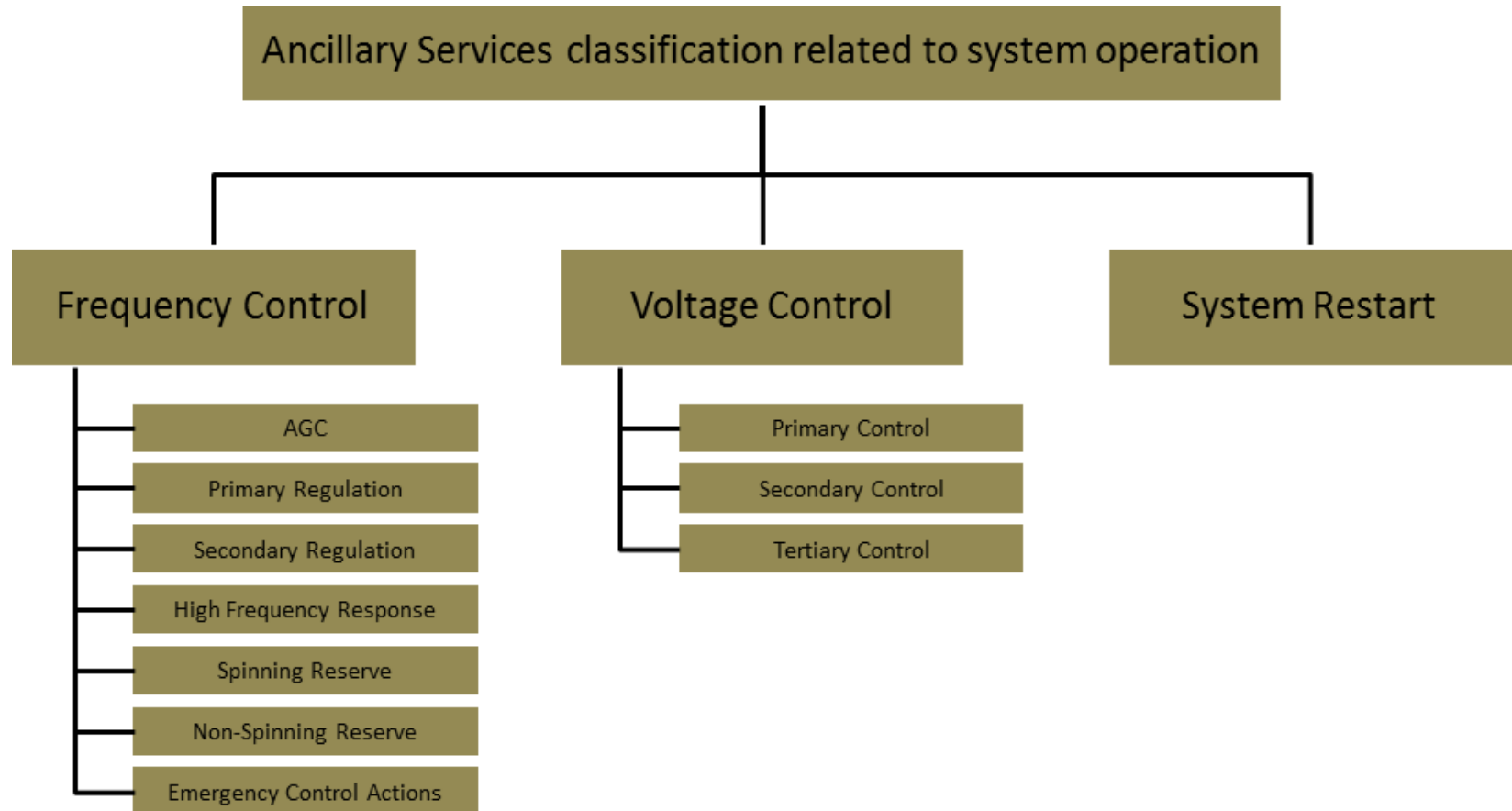
- Ancillary Services are necessary for the quality of the electricity supply
- The quality of electricity is tied to Security of supply
 - Frequency control
 - Voltage control
 - System restart
- These quality aspects are collective
- Collective goods (also: public goods) cannot be left to the market alone
- The System Operator is responsible for the quality and therefore for the provision of ancillary services

Security of Supply – Electropedia

Section 191-21: Electric power system related concepts

- security (of an electric power system)
 - Ability of an electric power system to operate in such a way that credible events do not give rise to loss of load, stresses of system components beyond their ratings, bus voltages or system frequency outside tolerances, instability, voltage collapse, or cascading
 - Note 1 – This ability may be measured by one or several appropriate indices.
 - Note 2 – This concept is normally applied to bulk power systems.
 - Note 3 – In North America, this concept is usually defined with reference to instability, voltage collapse and cascading only.
- Security of the electricity supply
 - Ability of an electric power system to provide electric power and energy to end-users with evaluation of existing standards and contractual agreements at the point of supply

General classification of ancillary services



We will focus on frequency-related services in the following, as they directly relate to system balance

Ancillary services: Categories

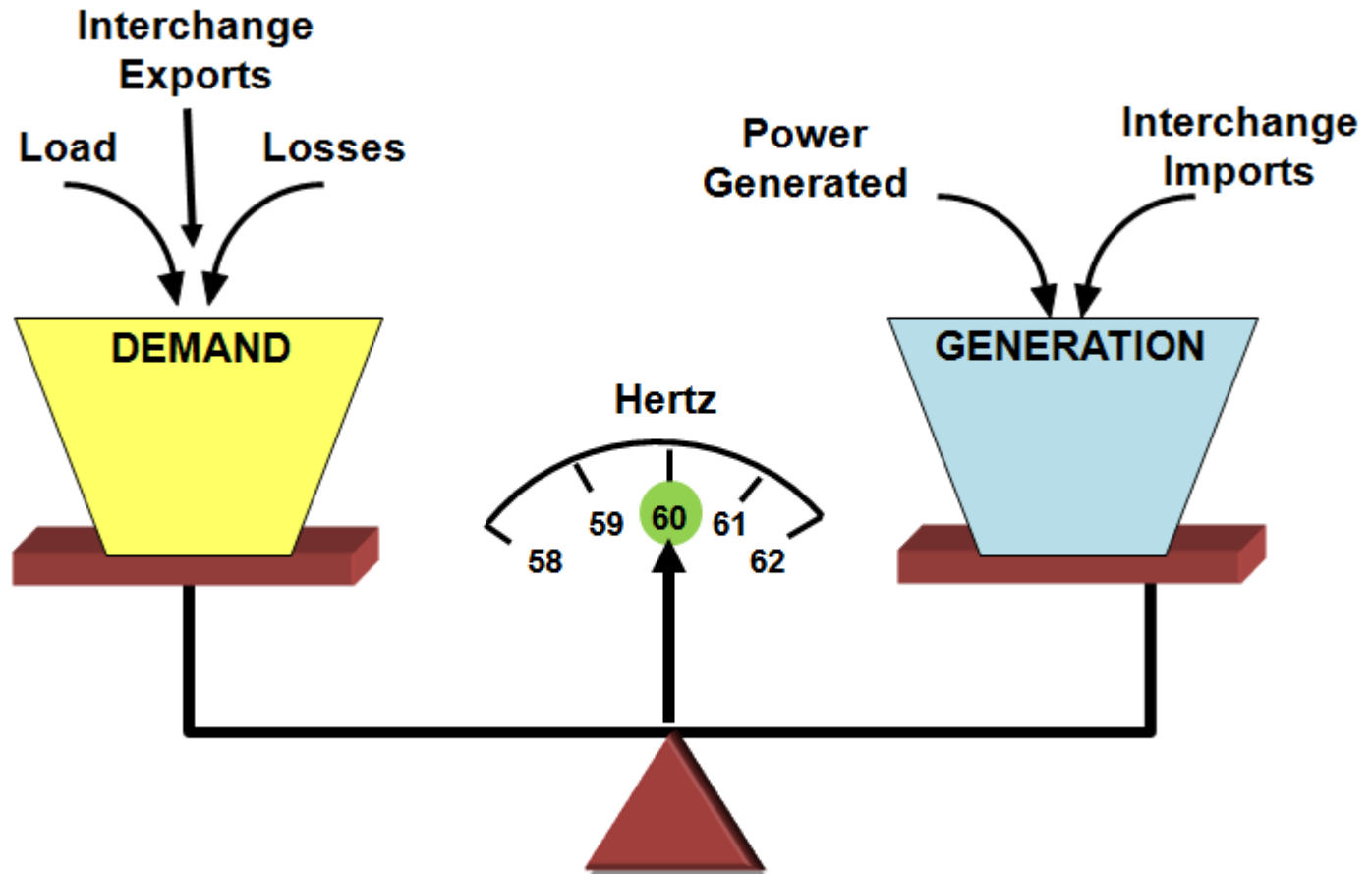
- Three main categories of ancillary services are
 - 1) **Frequency-supporting services** that enable the TSO to maintain the instantaneous balance between power supply and demand, and therefore preserve the frequency of the system in 50 Hz (or 60 Hz in the U.S.),
 - 2) **Voltage-supporting services** that enable the TSO to preserve the voltage stability of the system (generally by controlling the supply of reactive power),
 - 3) **Black-start services** that enable the TSO to re-start the system after a blackout.
- In this lecture, we focus on frequency-supporting services only!

On the need for frequency-supporting services:

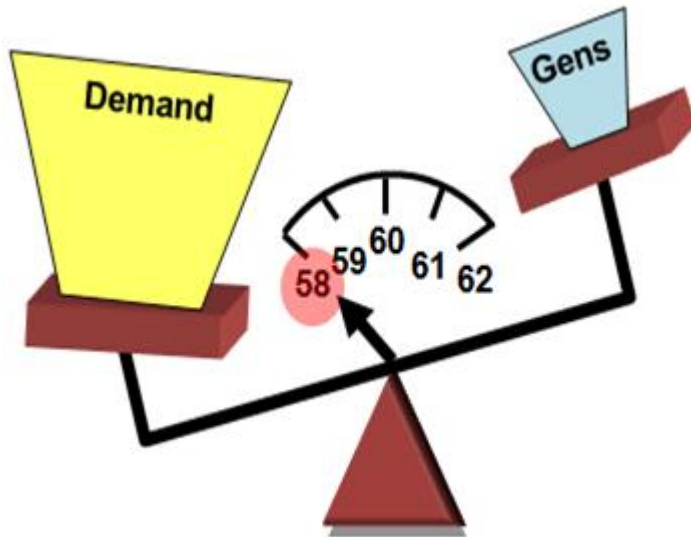
We can almost always expect a mismatch between “scheduled” power supply and demand due to the following disturbances:

- Forecast errors in the prediction of renewable power supply and in the prediction of demand,
- Potential outages, i.e., failure of a generator, or a transmission line, or a transformer, or any other power system asset.

Balance management



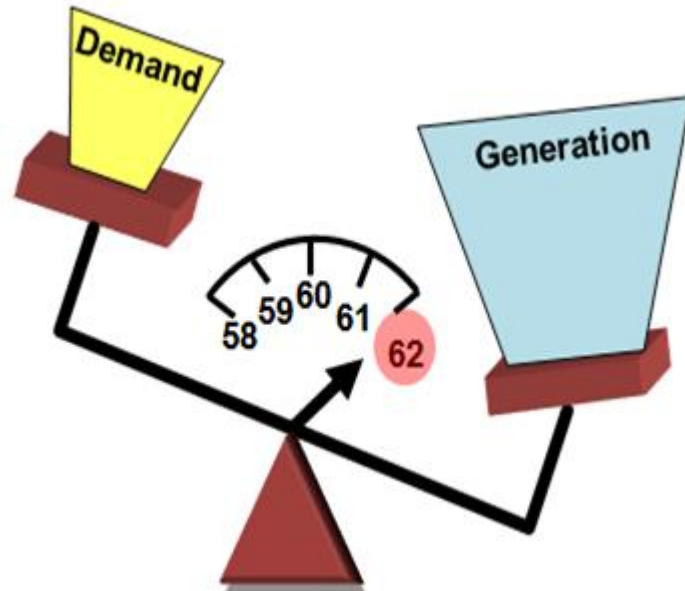
Imbalance



Total generation < total demand

Frequency decrease

Generators (momentarily) slow down

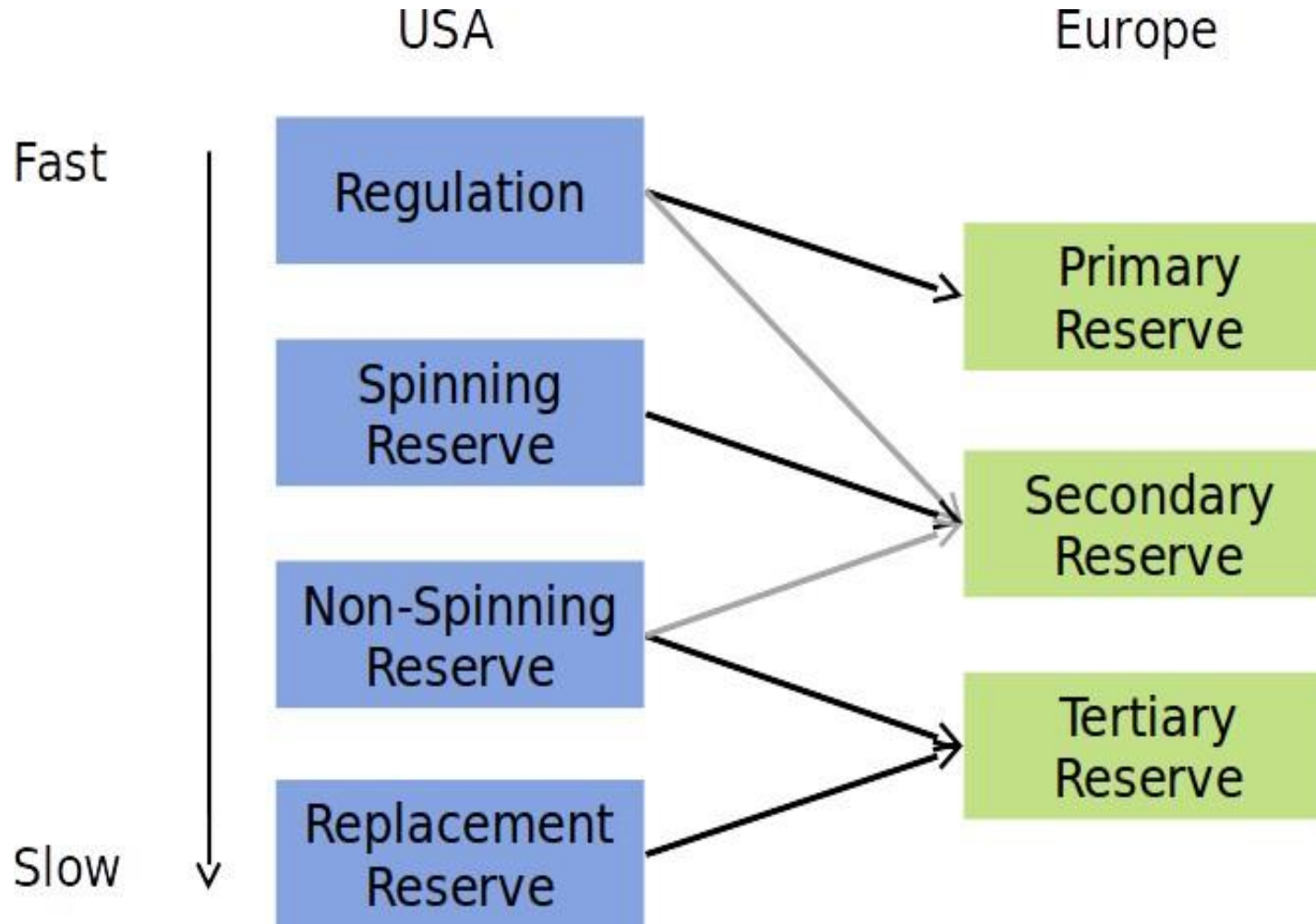


Total generation > total demand

Frequency increase

Generators (momentarily) speed up

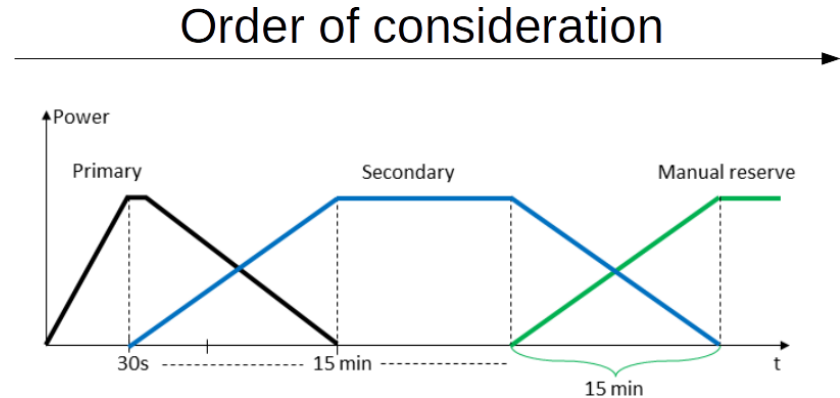
Beware of naming conventions



Activation sequence

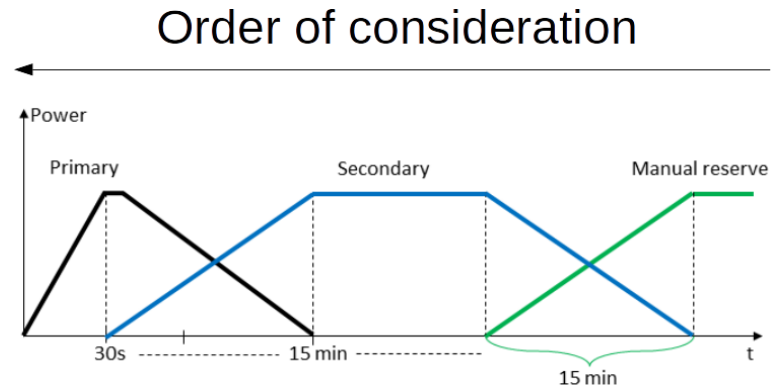
Two main approaches:

❑ Reactive (/corrective) approach



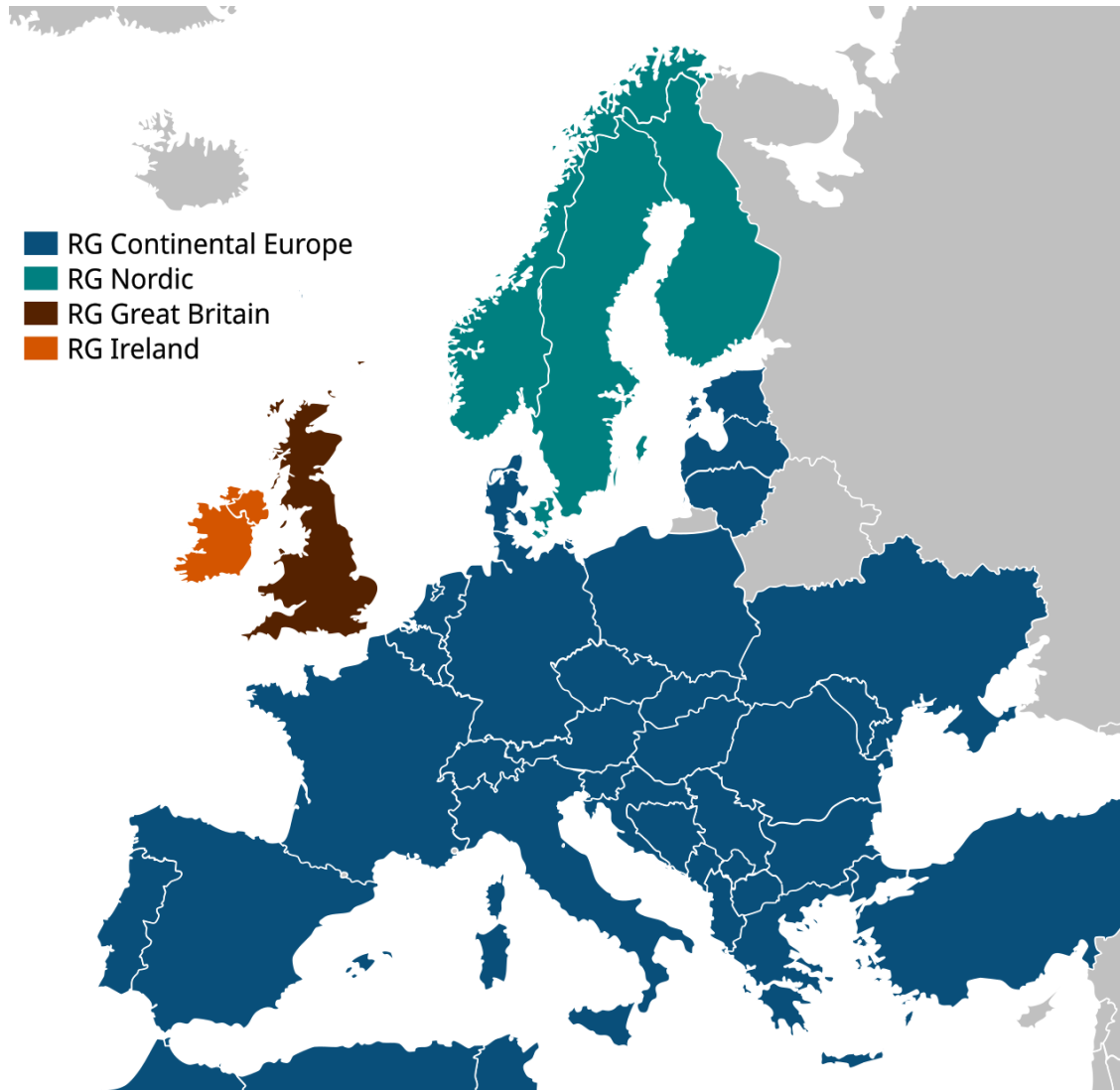
A disturbance occurs. TSO starts coping with it by activating primary reserve, secondary reserve and eventually manual reserve.

❑ Proactive (/preventive) approach

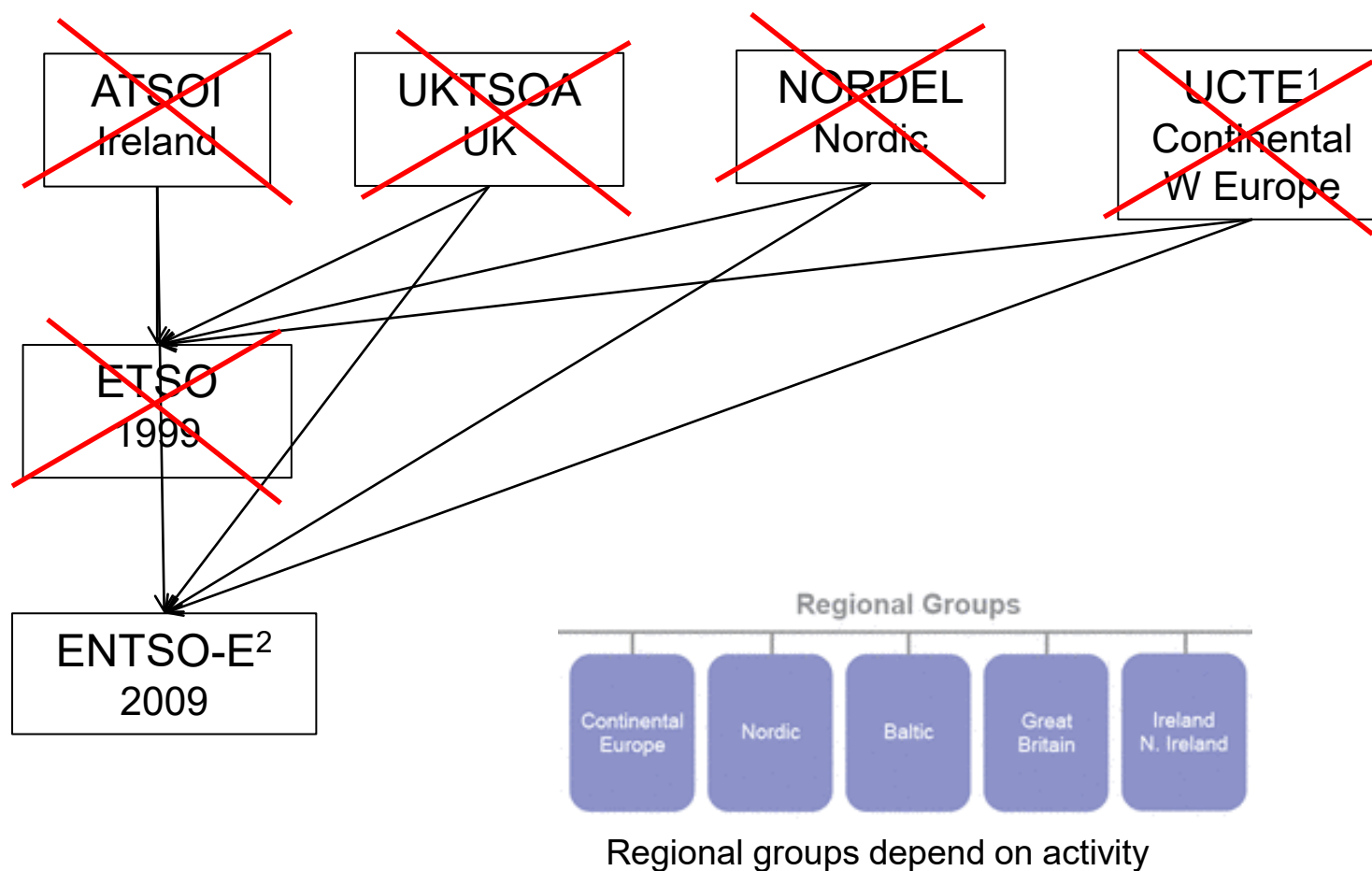


TSO anticipates a disturbance and activates manual reserve to be ready in 30-60 minutes. Closer to the disturbance (5-10 minutes ahead), she/he then activates secondary reserve and eventually primary reserve to be used in the real-time.

Synchronous grid areas in Europe



International Cooperation



Harmonization & Integration

- Before 1999 rules were mainly determined on a national basis and coordinated within UCTE, Nordel etc.
- Significant differences in approaches and nomenclature
- In 2007 ETSO published a Reference Model¹
- Terminology from Reference Model is now used in the development of new common Grid Codes for **ENTSO-E**
- Reason: integration of European electricity markets

¹ 4th ETSO Report on Balance Management Harmonization and Integration, January 2007

Reference Model



Fast

slow

FFR (Nordic)

FCR

FRR

RR

Fast
Frequency
Reserve

Frequency
Containment
Reserve

Frequency
Restoration
Reserve

Replacement
Reserves

automatic
activation

Capacity market

automatic
activation

Capacity market

automatic
activation

Capacity
market

Energy
market

manual
activation

Capacity
market

Energy
market

manual
activation

Capacity
market
(optional)

Energy
market

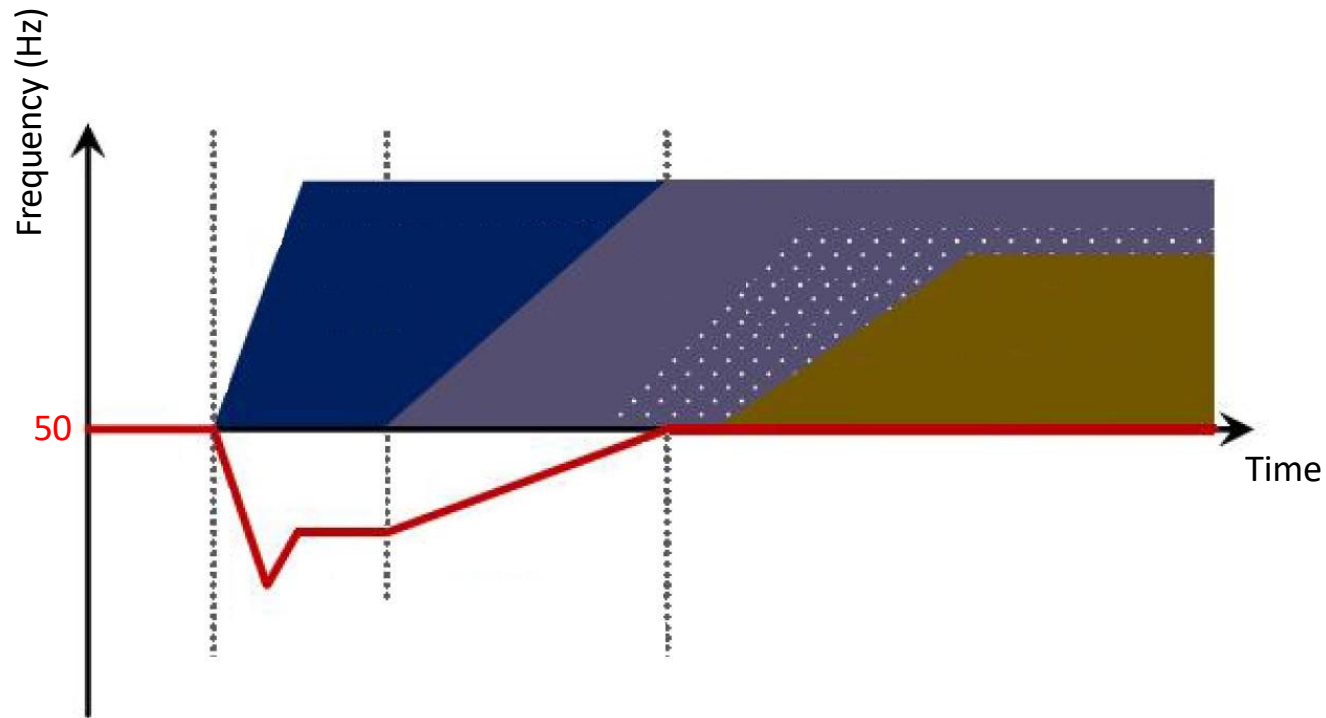
inertia-like response (sub-second)

stabilizes frequency

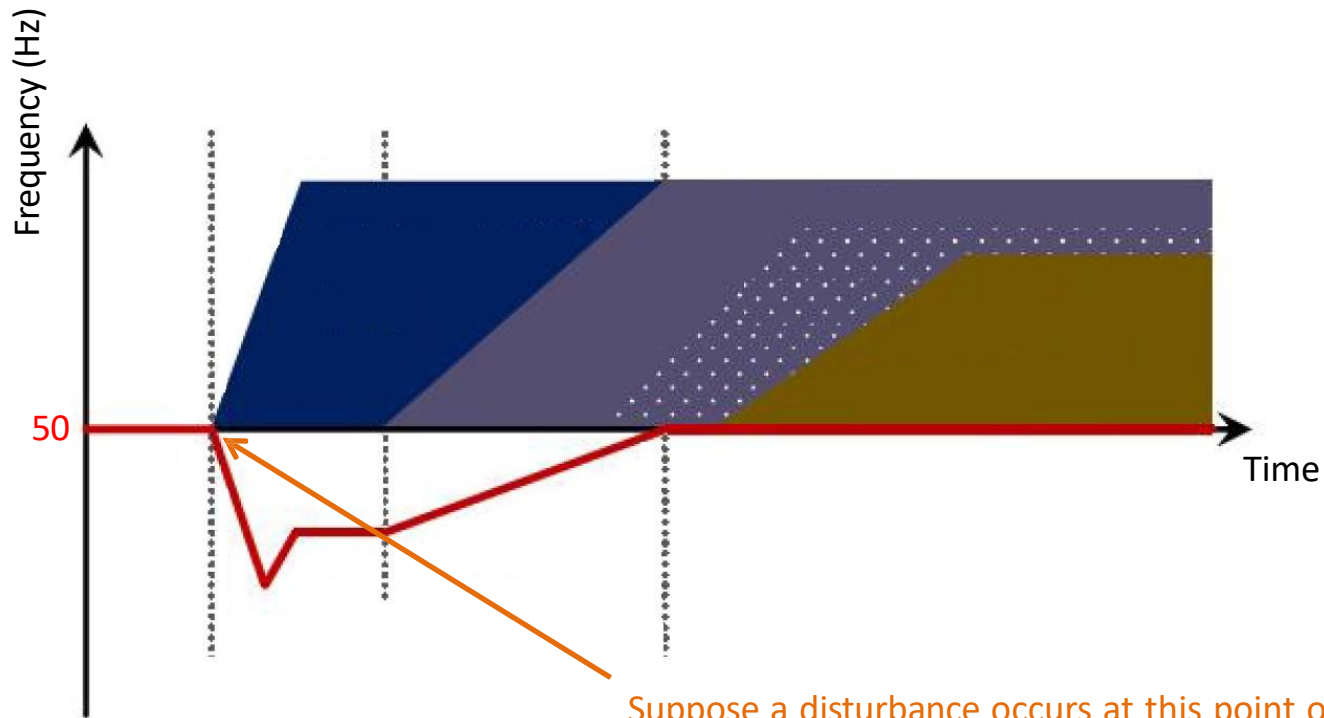
restores frequency

long-term replacement (less visible)

Frequency-supporting ancillary services

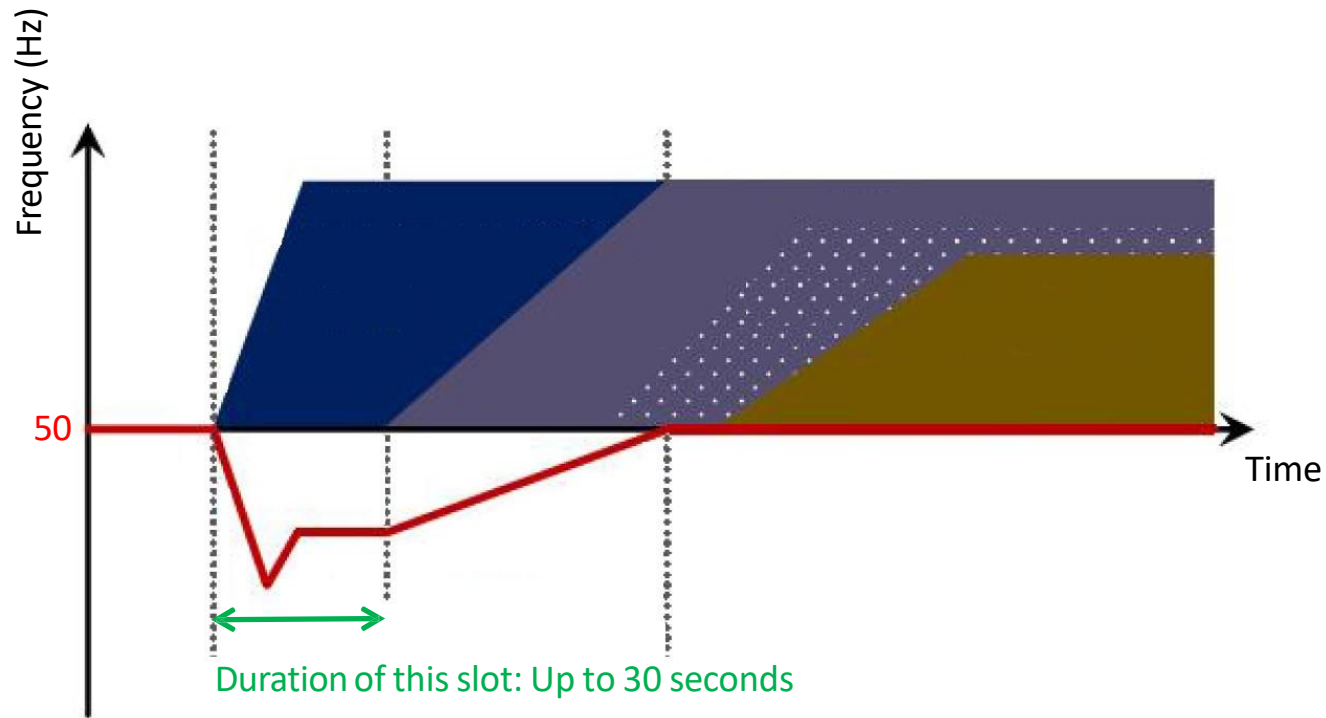


Frequency-supporting ancillary services



Suppose a disturbance occurs at this point of time, leading to an imbalance (in this case, it is a power deficit, causing a frequency drop)

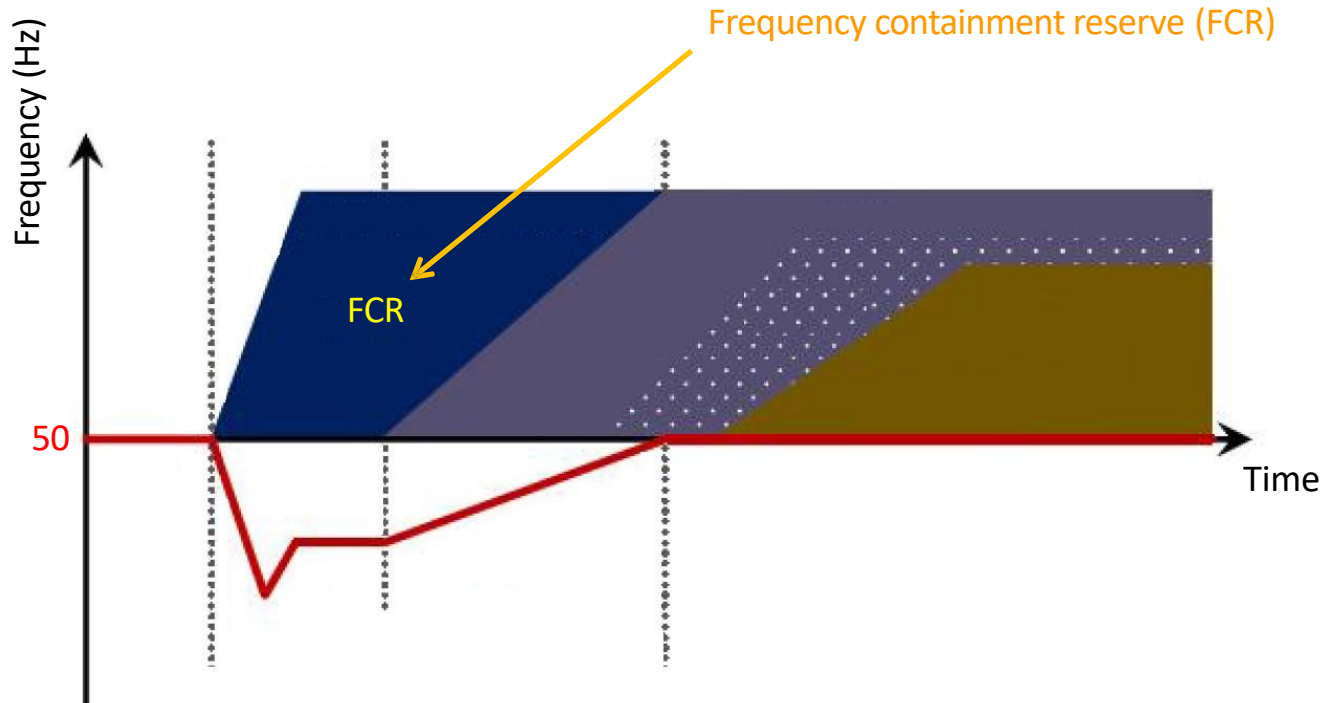
Frequency-supporting ancillary services



Duration of this slot: Up to 30 seconds

Goal: frequency containment process (to avoid this drop becomes even worse)

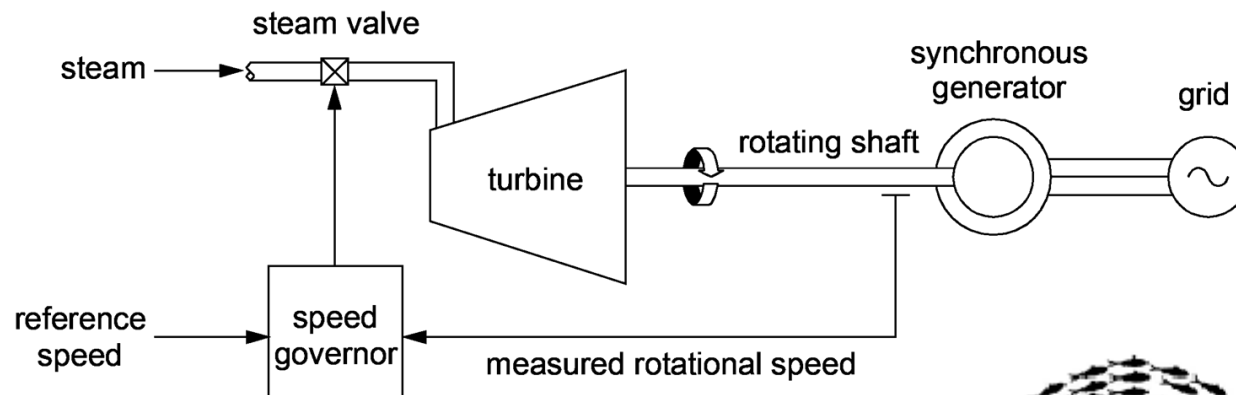
Frequency-supporting ancillary services



- FCR is the fastest frequency-supporting ancillary service!
- Reaction and full response depend on the FCR provider, but it could be immediate or may take a few seconds.
- It is also known as “**primary reserve**” and “**regulating reserve**”.

Frequency Containment Reserves (FCR)

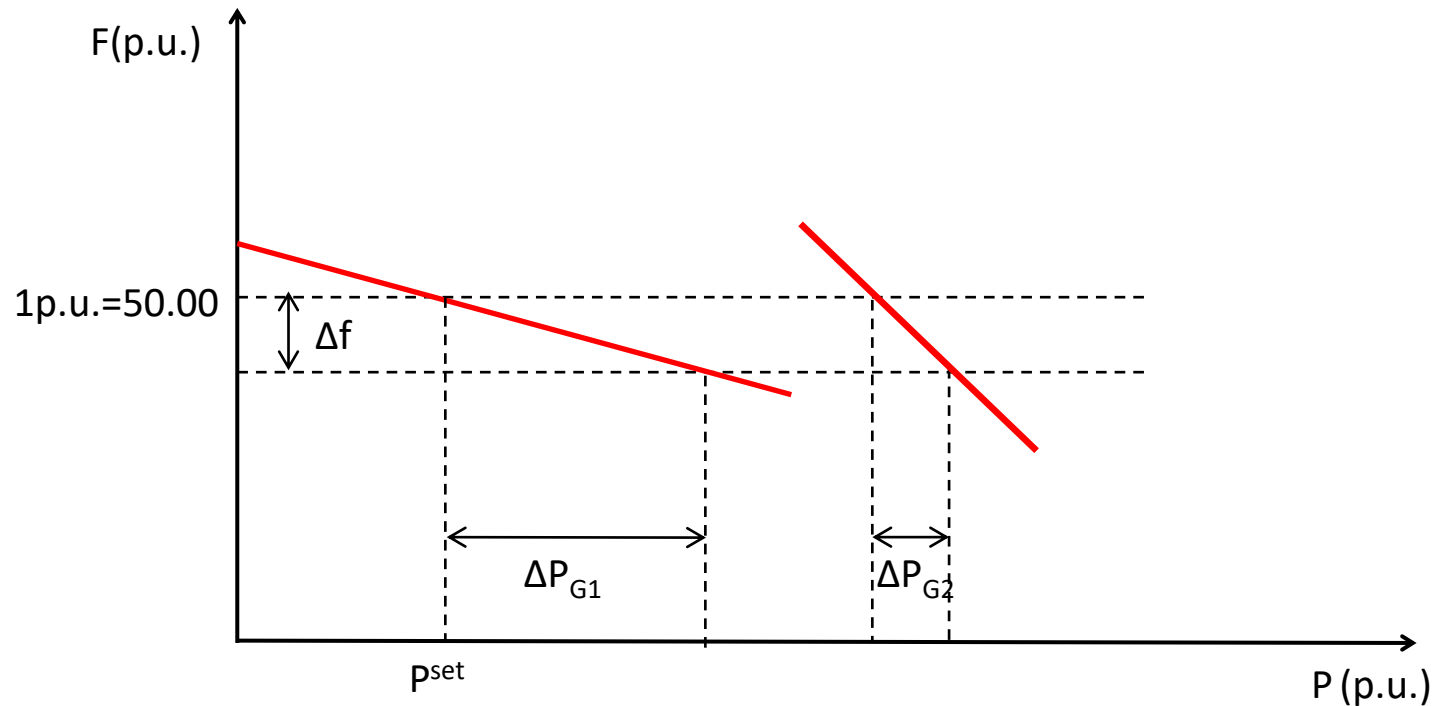
- Stabilize system frequency after disturbance or incident
- Time frame of seconds
- Does not restore frequency to set point
- Activated locally and automatically
- Provided by generation (primarily), controllable load and HVDC interconnections



FCR – unit droop (statikk in Norwegian)

- Unit droop defined as: $s = -\frac{\Delta f / f_n}{\Delta P / P_n}$ in %
- This means that $\frac{\Delta P}{P_n} = -\frac{1}{50} \cdot \frac{\Delta f}{s} = -0.02 \cdot \frac{\Delta f}{s}$
- So, for $\Delta f = -0.1$ Hz:
 - $s = 4$ % means that the unit will increase output with 5 %
 - $s = 8$ % means that the unit will increase output with 2,5 %
- The lower s , the stronger the unit's contribution

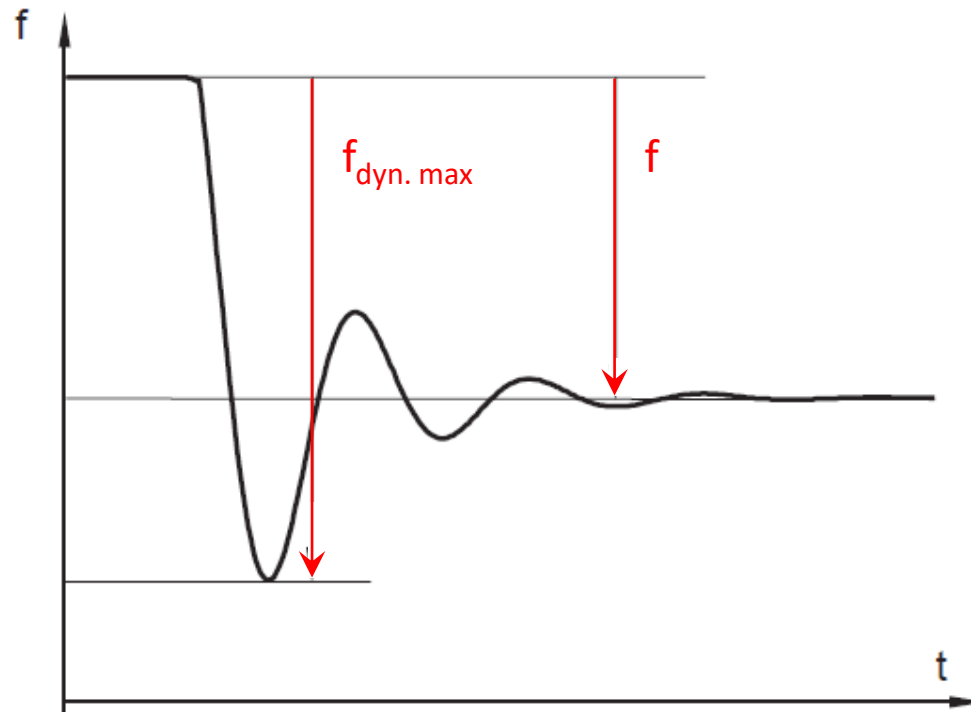
Unit droop illustration (Load-frequency characteristics)



Steeper line $\rightarrow$ higher droop and less contribution

$$\Delta P_{G2} < \Delta P_{G1}$$

FCR: transient and steady state characteristic



$f_{\text{dyn. max.}}$ = Dynamic frequency deviation

f = Quasi-steady-state deviation

Source; UCTE Operation Handbook Policy 1 Appendix 1: Load-Frequency Control and Performance ([LINK](#))

FCR Parameters in Continental Europe and Nordic system

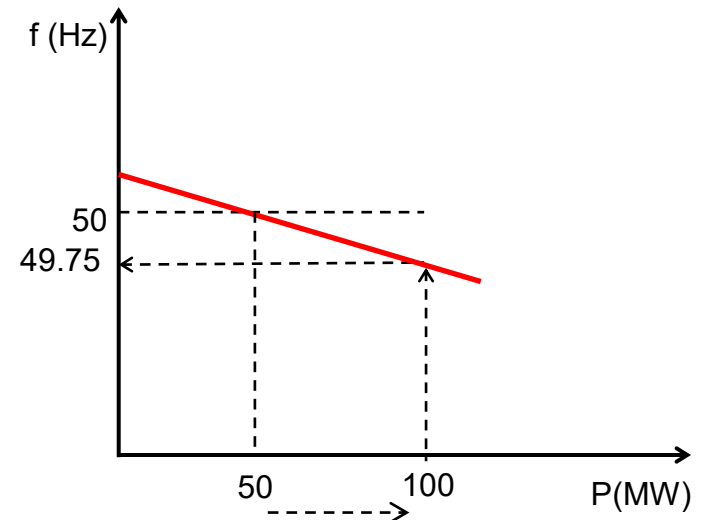
	Continental Europe	Nordic
Standard frequency deviation range	± 50 mHz	± 100 mHz
Maximum frequency deviation	± 800 mHz	± 500 mHz
Maximum quasi-steady-state frequency deviation	± 200 mHz	± 500 mHz
Reference incident	3000 MW	Biggest unit in operation, calculated weekly
Tolerance range for FCR activation	± 20 mHz	-
Full activation time of FCR	30s	3 min (normal, FCR-N) 30 s (disturbance, FCR-D)

A Standalone Generator

A standalone generator feeding a single load supplies 50 MW at 50 Hz. The generator has a nominal rated power of 200 MW and its speed governor has a droop of 2% p.u. The active power consumption of the load suddenly increases to 100 MW. What is the frequency drop caused by this event?

$$\Delta P = 100 - 50 = 50 \text{ MW}$$

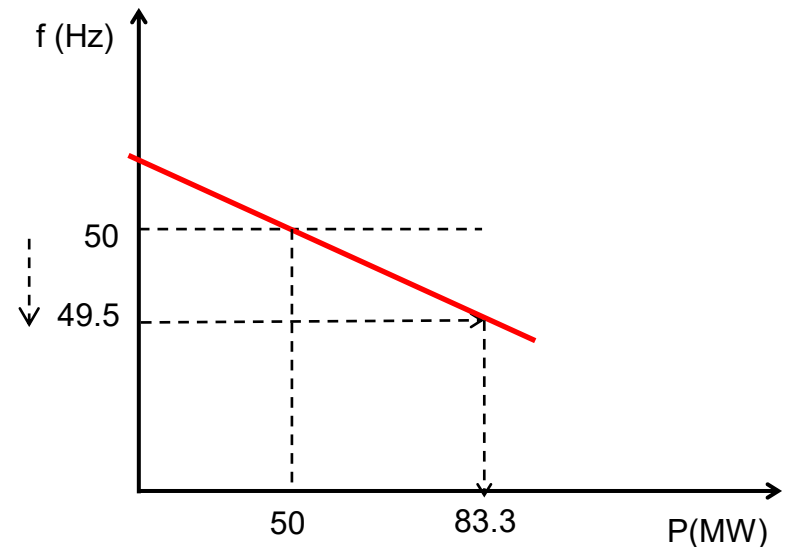
$$s = -\frac{\frac{\Delta f}{f_n}}{\frac{\Delta P}{P_n}}, \Delta f = -s \cdot f_n \left(\frac{\Delta P}{P_n} \right) = -0.02 \times 50 \times \left(\frac{50}{200} \right) = -0.25 \text{ Hz}$$



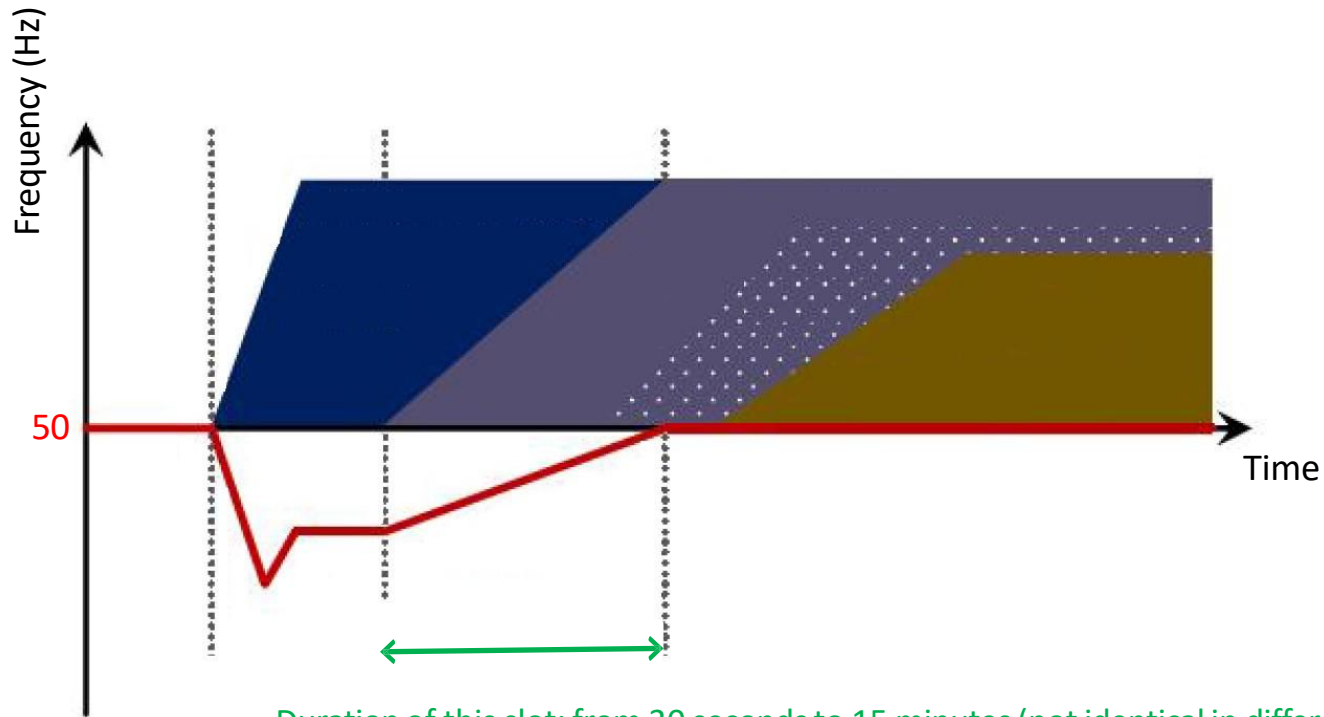
A Generator Connected to an Infinite Bus

A generator connected to an infinite bus supplies 50 MW at the nominal rated frequency of 50 Hz. The generator has a nominal rated power of 200 MW and its speed governor has a droop of 6% p.u. When the system frequency drops to a value of 49.5 Hz, what is the increased active power supplied by the generator?

$$s = -\frac{\frac{\Delta f}{f_n}}{\frac{\Delta P}{P_n}}, \Delta P = -\frac{\Delta f}{s} \cdot \frac{P_n}{f_n} = \frac{0.5}{0.06} \times \frac{200}{50} = 33.3 \text{ MW}$$



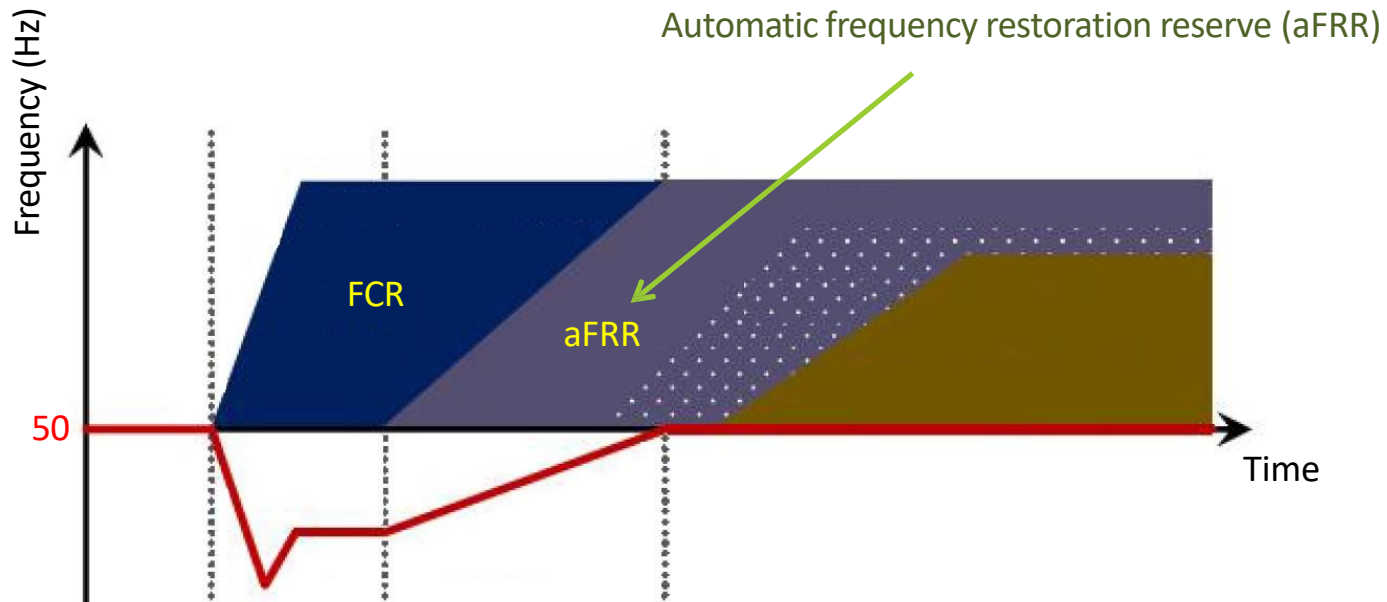
Frequency-supporting ancillary services



Duration of this slot: from 30 seconds to 15 minutes (not identical in different countries)

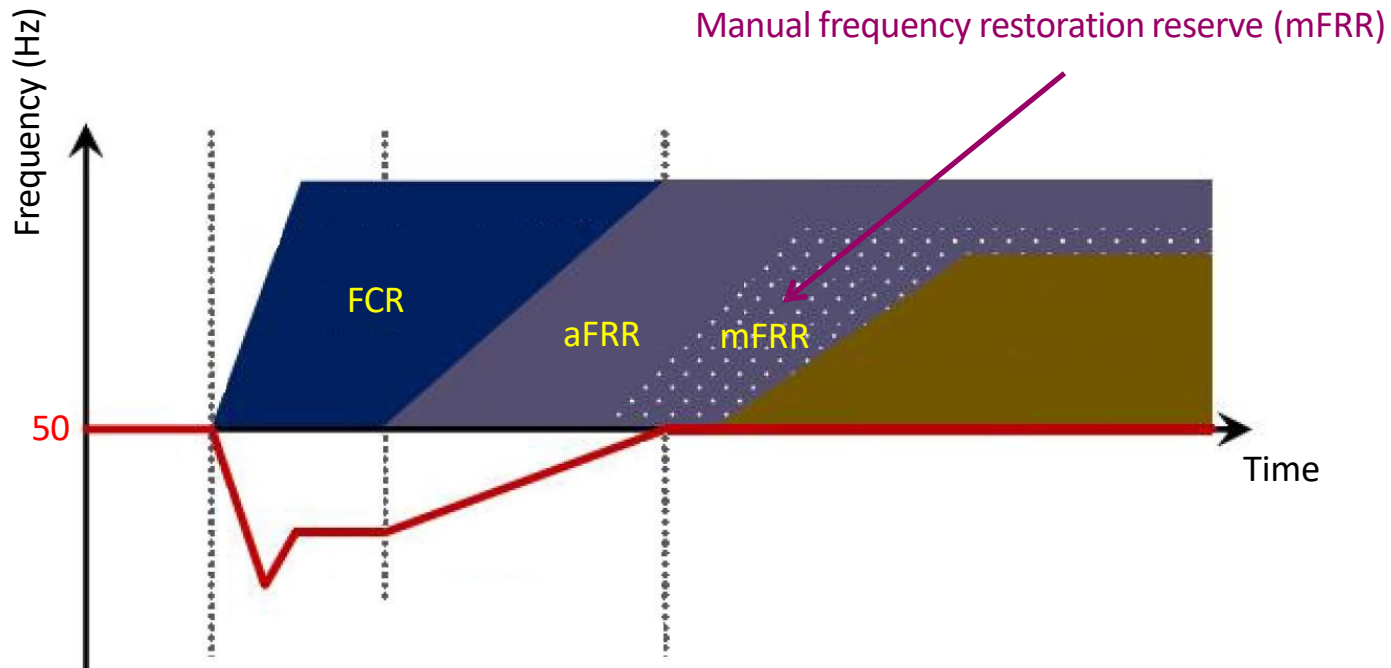
Goal: frequency restoration process

Frequency-supporting ancillary services



- Reaction in a few seconds, but full response within 5-15 minutes
- Potential aFRR providers: spinning (on-line) generators, and non-spinning (off-line) generators if they are fast-start enough
- It is also known as “**secondary reserve**” and “**operating reserve**”.

Frequency-supporting ancillary services



- Full response within 15-60 minutes (this is a comparatively slow frequency-supporting ancillary service)
- It is also known as “**tertiary reserve**” and “**replacement reserve**”.

Frequency Restoration Reserves (FRR)

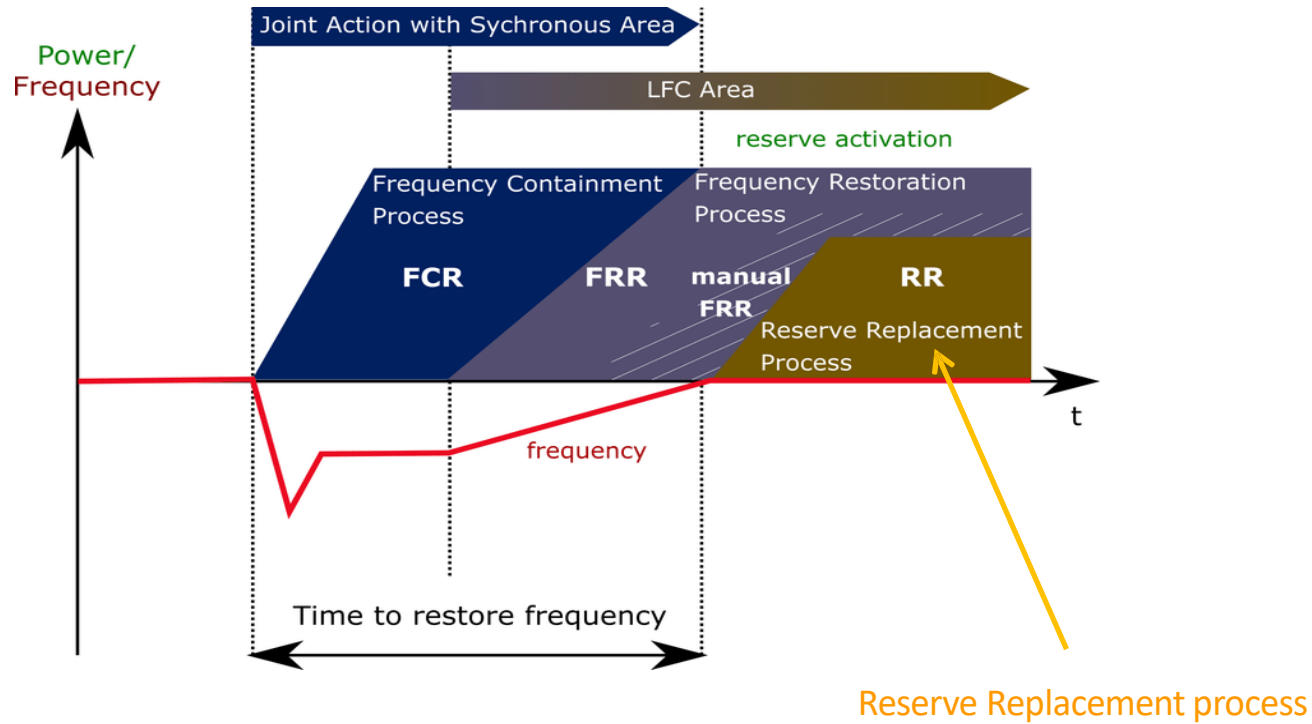
- FRR aim to restore frequency back to set-point
- Modifies controller set-points of FCR
- Example:
 - Unit set to produce 100 MW
 - Droop 4 %
 - Frequency falls due to disturbance
 - Unit output increases
 - Frequency stabilizes at 49.95 Hz, unit produces 102.5 MW
 - Unit set-point adjusted to produce 102.5 MW at $f = 50$ Hz
- FRR can be automatic or manual (aFRR) or (mFRR)
- Time frame 15 minutes



Why are FRR needed?

- Restore frequency to the nominal value
- But at least as important: to release FCR!
- What happens if FCR is exhausted and a new event occurs?

Frequency-supporting ancillary services

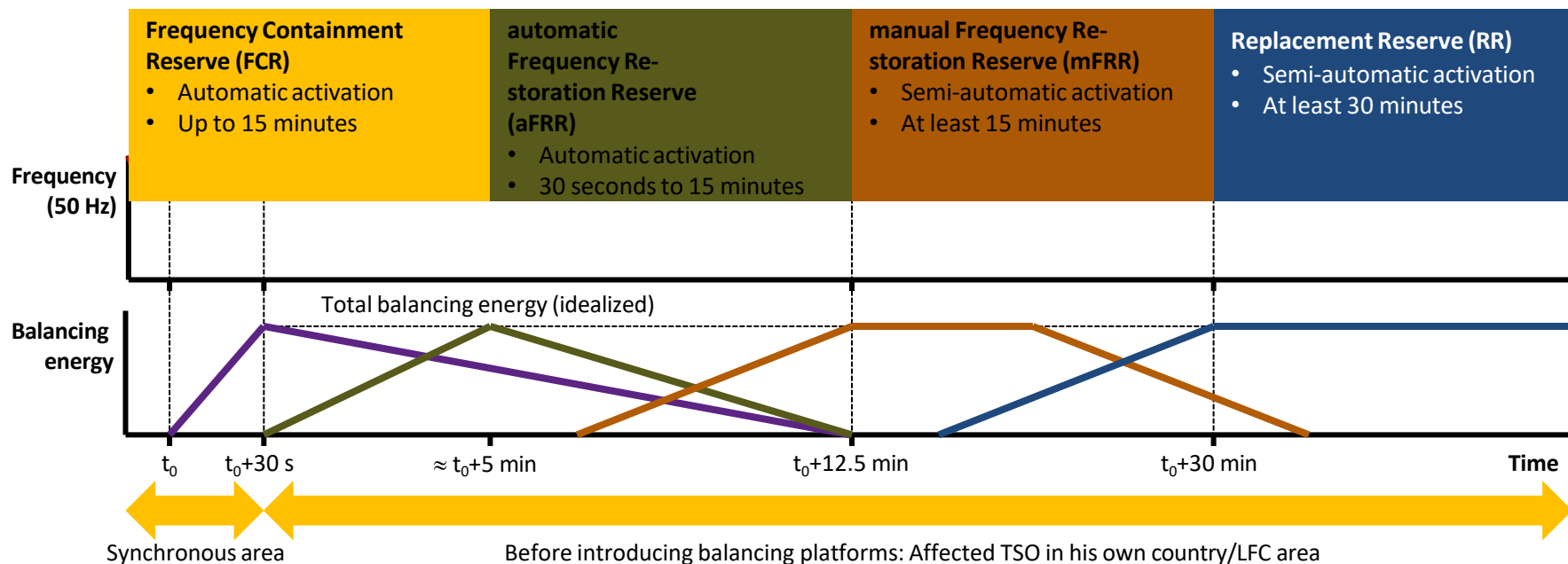


Replacement Reserves (RR)

- Prepare for further imbalances in case FCR / FRR has already been activated up to a certain extent
- Manual and central activation
- Units made explicitly available for this purpose
 - independent of FCR and FRR

Overview of Frequency Services

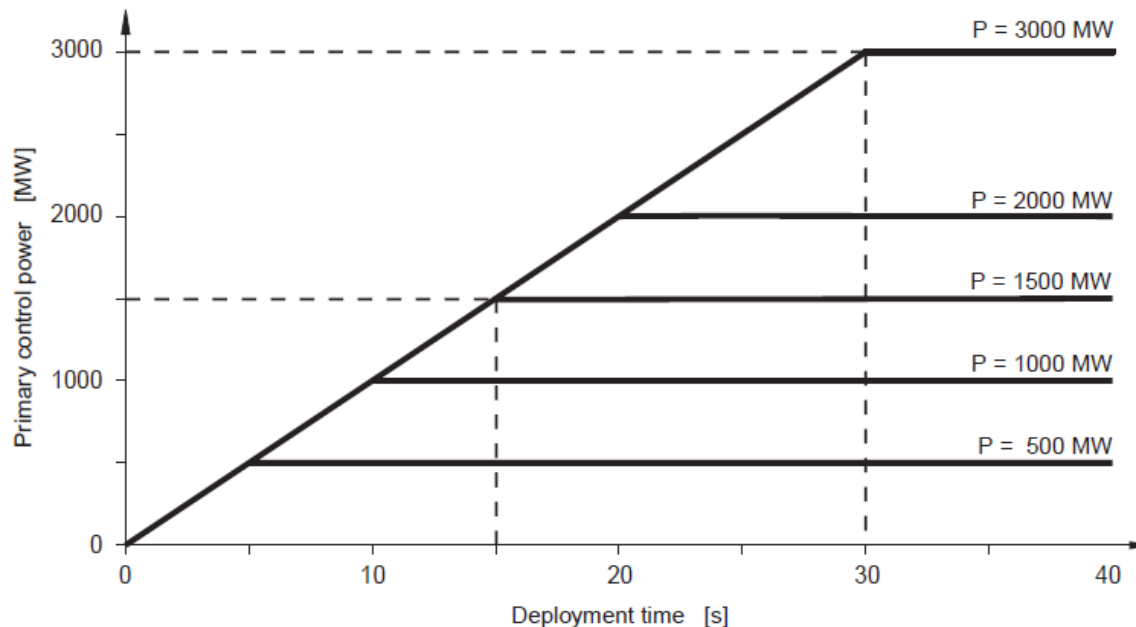
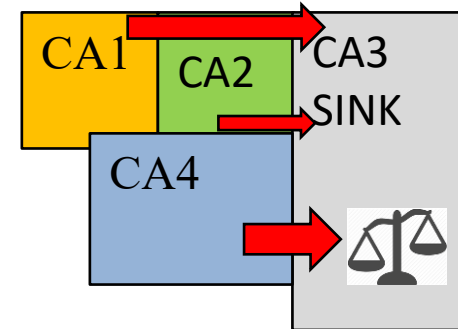
- Balancing services according to activation time (without imbalance netting, IN)



- Different balancing services allow TSOs to cover imbalances in different time frames (from few seconds to various quarter hours) and areas (from LFC areas to synchronous areas).

RGCE – FCR

- Primary Control Reserve
 - 3000 MW for whole system (2 largest units)
 - dead band ± 20 mHz
 - fully activated at ± 200 mHz
 - 50 % within 15 seconds, 100 % within 30 seconds
- Instantaneous maximum frequency deviation ± 800 mHz



RGCE – aFRR (ACE)

- Secondary Control Reserve
- Based on Area Control Error (ACE)

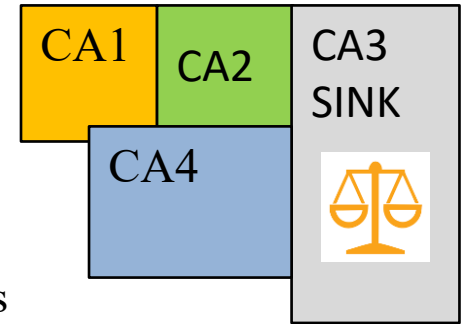
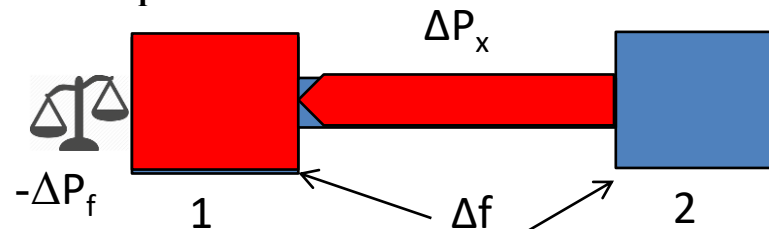
$$ACE = \Delta P_{xi} + R_i \cdot \Delta f$$

where ΔP_{xi} is deviation in power exchange with other areas

- Frequency bias for area: $R = -\frac{\sum_j \Delta P_j}{\Delta f}$
 - sum of contribution of all units + load contribution
- When everything is normal, $ACE = 0$
- Two areas, unit outage ΔP_f in Area 1:

$$\begin{aligned} ACE_1 &= \Delta P_{x1} + R_1 \cdot \Delta f = \Delta P_x + R_1 \cdot \Delta f \\ &= R_2 \cdot \Delta f + R_1 \cdot \Delta f \\ &= (R_2 + R_1) \cdot \Delta f = -\Delta P_f \end{aligned}$$

$$\begin{aligned} ACE_2 &= \Delta P_{x2} + R_2 \cdot \Delta f = -\Delta P_x + R_2 \cdot \Delta f \\ &= -R_2 \Delta f + R_2 \cdot \Delta f = 0 \end{aligned}$$



Frequency Bias (The network power frequency characteristic)

Consider two generators operating in parallel in a given control area. Generator 1 has a nominal rated power 200 MW and its speed governor has a droop 2% p.u. Generator 2 has a nominal rated power 200 MW and its speed governor has a droop 6% p.u. What is the frequency bias of this control area? How much power is required to correct the frequency drop to half a Hertz?

$$\left. \begin{aligned} R &= -\frac{\Delta P}{\Delta f} = -\frac{\sum_i \Delta P_i}{\Delta f} \\ s_i &= -\frac{\frac{\Delta f}{f_n}}{\frac{\Delta P_i}{P_{in}}}, \Delta P_i = -\frac{\Delta f}{s_i} \cdot \frac{P_{in}}{f_n} \end{aligned} \right\} \Rightarrow R = -\frac{\sum_i -\frac{\Delta f}{s_i} \cdot \frac{P_{in}}{f_n}}{\Delta f} = \sum_i \frac{1}{s_i} \cdot \frac{P_{in}}{f_n}$$

$$R = \frac{1}{0.02} \times \frac{200}{50} + \frac{1}{0.06} \times \frac{200}{50} = 266.6 \frac{MW}{Hz}$$

$$\Delta f = -0.5 Hz \text{ \& } R = 266.6 \frac{MW}{Hz}, R = -\frac{\Delta P}{\Delta f} \Rightarrow \Delta P = -R \cdot \Delta f = 266.6 \times 0.5 = 133.3 MW$$

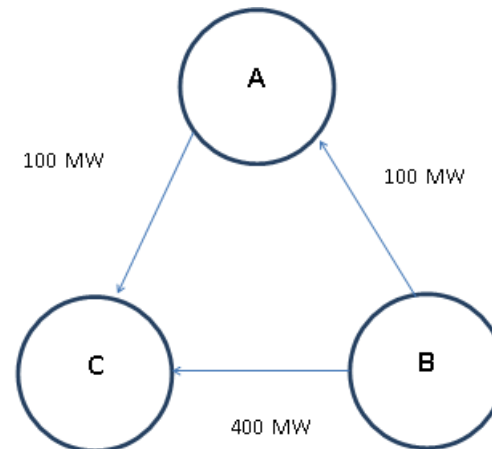
Area Control Error (ACE)

Consider a network that is divided into three control areas as shown below. Control area C imports a scheduled amount of power (500 MW) from control area B; note that 100 MW flows via the intertie connecting with area A, which has a zero-level scheduled import/export.

The frequency bias of the three control areas:

- Area A : $R_A = 13000$ MW/Hz
- Area B : $R_B = 16000$ MW/Hz
- Area C : $R_C = 11000$ MW/Hz

When suddenly a 400 MW generator is lost by a forced outage in control area B, what is the frequency drop in the whole system? Calculate ACE values for each area.



Area Control Error (ACE)-Continued

- Frequency bias for the whole system: $R = R_A + R_B + R_C = 40\,000 \frac{MW}{Hz}$
- When suddenly a 400 MW generator is lost: $\Delta f = -\frac{\Delta P}{R} = -\frac{400}{40\,000} = -0.01 Hz$
- Governed by FCR, the generator in three areas increase their power output

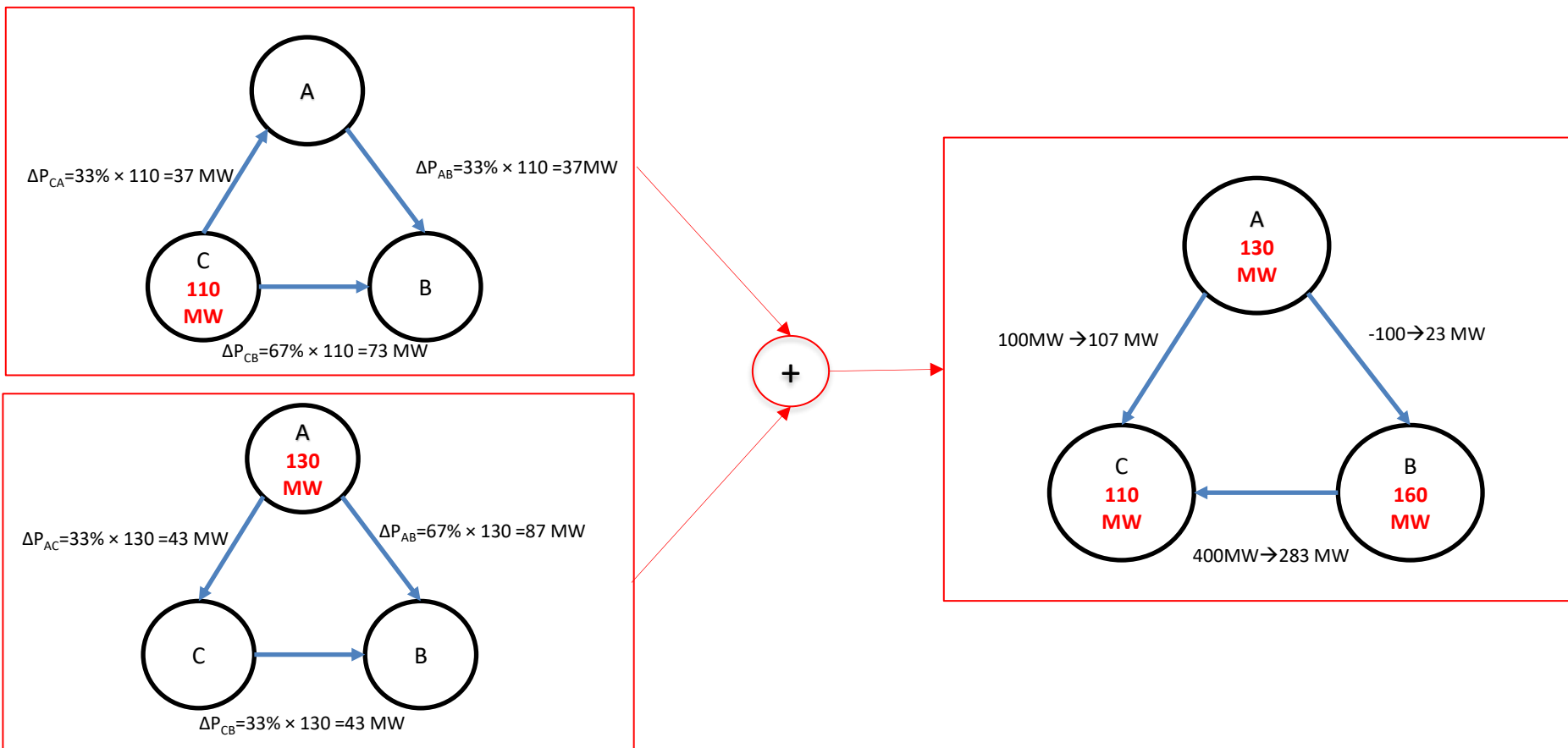
$$\Delta P_A = -R_A \cdot \Delta f = 13\,000 \times 0.01 = 130 MW$$

$$\Delta P_B = -R_B \cdot \Delta f = 16\,000 \times 0.01 = 160 MW$$

$$\Delta P_C = -R_C \cdot \Delta f = 11\,000 \times 0.01 = 110 MW$$

Area Control Error (ACE)-Continued

- This affects the power flow through intertie connection. From the original power flow, we can recognize that 67% of the exported power from **C** to **B** flows through direct intertie connection and 33% flows via **A** area.



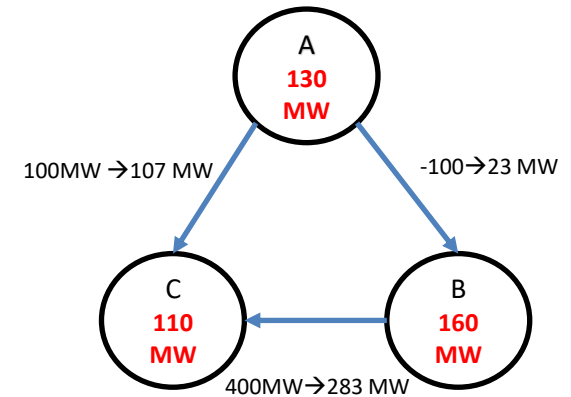
Area Control Error (ACE)-Continued

$$ACE = \Delta P_{ex} + R \cdot \Delta f$$

$$ACE_A = (0 - (-130)) + 13000 \times (-0.01) = 0 \text{ MW}$$

$$ACE_B = (-500 - (-260)) + 16000 \times (-0.01) = -240 - 160 = -400 \text{ MW}$$

$$ACE_C = (500 - 390) + 11000 \times (-0.01) = 0 \text{ MW}$$



Based on these ACE values, only FRR control actions have to be taken in area B.

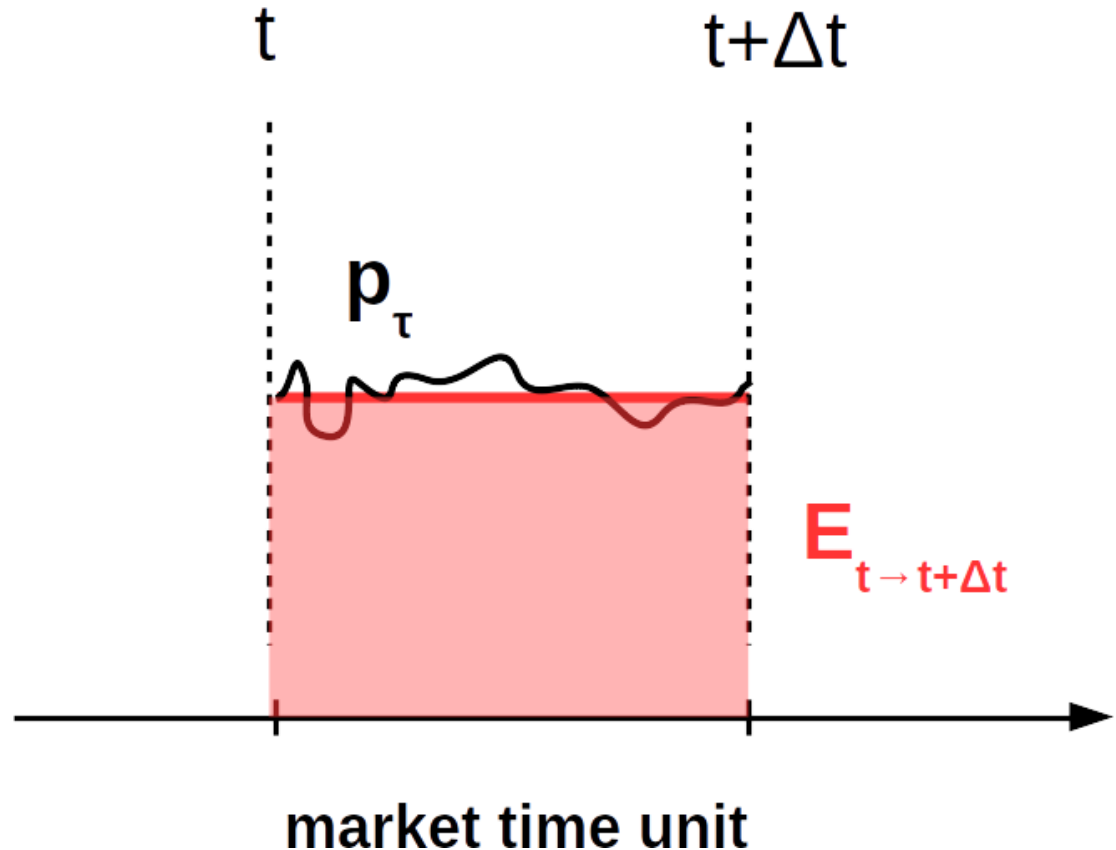
Payments for capacity and energy

From power to energy

Power is an instantaneous quantity (W), which we denote p_τ

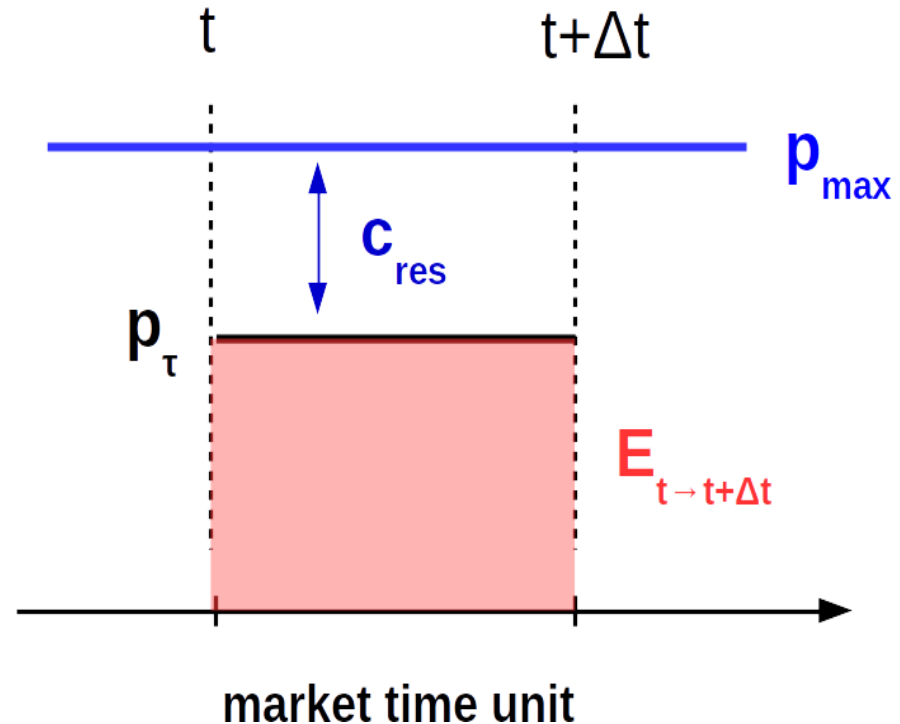
Energy is the result of integrating power generation over time (Wh), i.e.,

$$E_{t \rightarrow t+\Delta t} = \int_t^{t+\Delta t} p_\tau d\tau$$



Capacity and reservation

- At any time τ , power can be up to p_τ
- A part of that available power may be subject to *capacity reservation* (W), i.e.,
- $p_\tau \leq p_{max} - c_{res}, \quad \forall \tau$



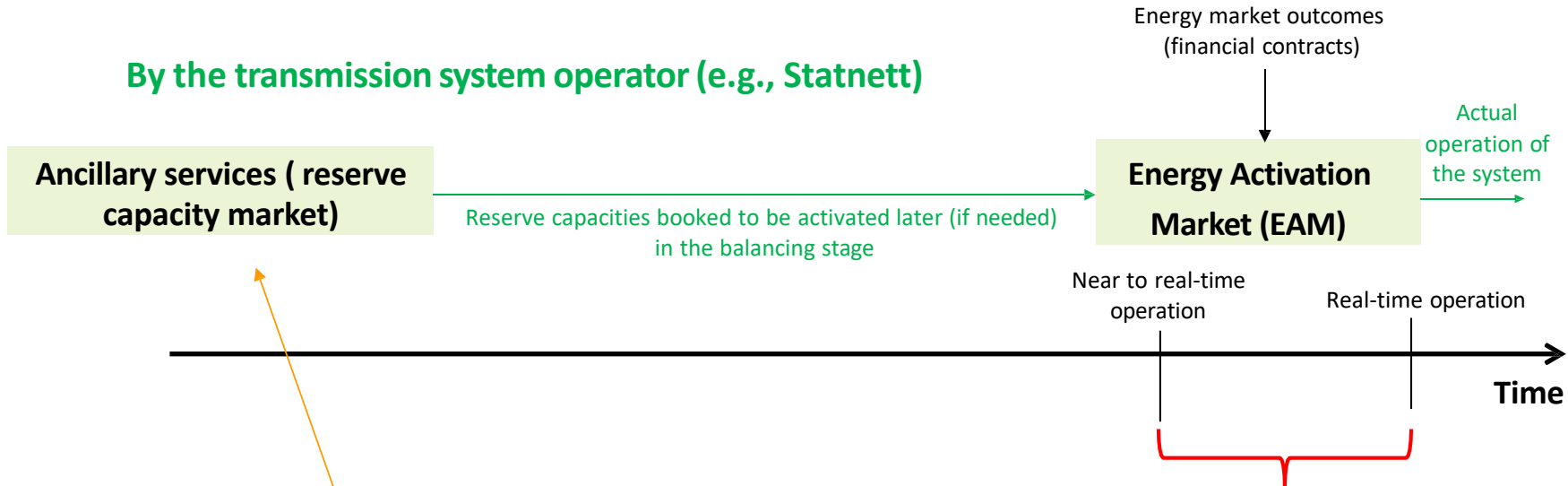
- Contracting of ancillary services by the system operator implies capacity reservation
- More generally capacity reservation can be in both up and down directions
- This hence affects the possibility to offer on the day-ahead electricity market

Provision of reserves

- Note the difference between reserve capacity
 - payment per MW for availabilityand balancing energy
 - payment for activated reserves during system operation
- Large differences between countries
- Reserve capacity:
 - Bilateral agreements between TSOs and generators
 - Yearly – monthly – weekly – daily auctions
- Tendency towards more market based mechanisms

Ancillary service markets

By the transmission system operator (e.g., Statnett)



TSO needs to “reserve” enough **capacity** for each ancillary service through a financial market, i.e., a reserve market to be cleared in advance, e.g., before the day-ahead electricity market. Another alternative for the TSO is to sign long-term bilateral contracts with service providers (non-market solution).

If needed, different types of ancillary services are **activated** by the TSO either in the real-time operation (in the case of FCR) or in this time slot (in the case of aFRR and mFRR)!

Question:

How to ensure that **sufficient** capacity for various ancillary services would be available in this time slot?

Payment for capacity and energy

Imagine that “Company A” can generate up to $p_{max}=100$ MW for a market time unit of interest.

3 illustrative cases:

- “Company A” gets a capacity reservation of 10MW for **upward** regulation, it can then only offer between 0 and 90MWh on the day-ahead electricity market
- Say capacity is reserved at 100 NOK/MW and all energy sold at $\lambda^s=200$ NOK/MWh, the resulting revenue is

$$\text{Revenue}(\text{“Company A”}) = 10 \times 100 + 90 \times 200 = 19000 \text{ NOK}$$

Payment for capacity and energy (2)

3 illustrative cases (cont.):

- “Company A” gets a capacity reservation of 10MW for **downward** regulation, it can then only offer between 10 and 100MWh on the day-ahead electricity market
- Say capacity is reserved at 100NOK/MW and all energy sold at $\lambda^S = 200\text{NOK/MWh}$, the resulting revenue is

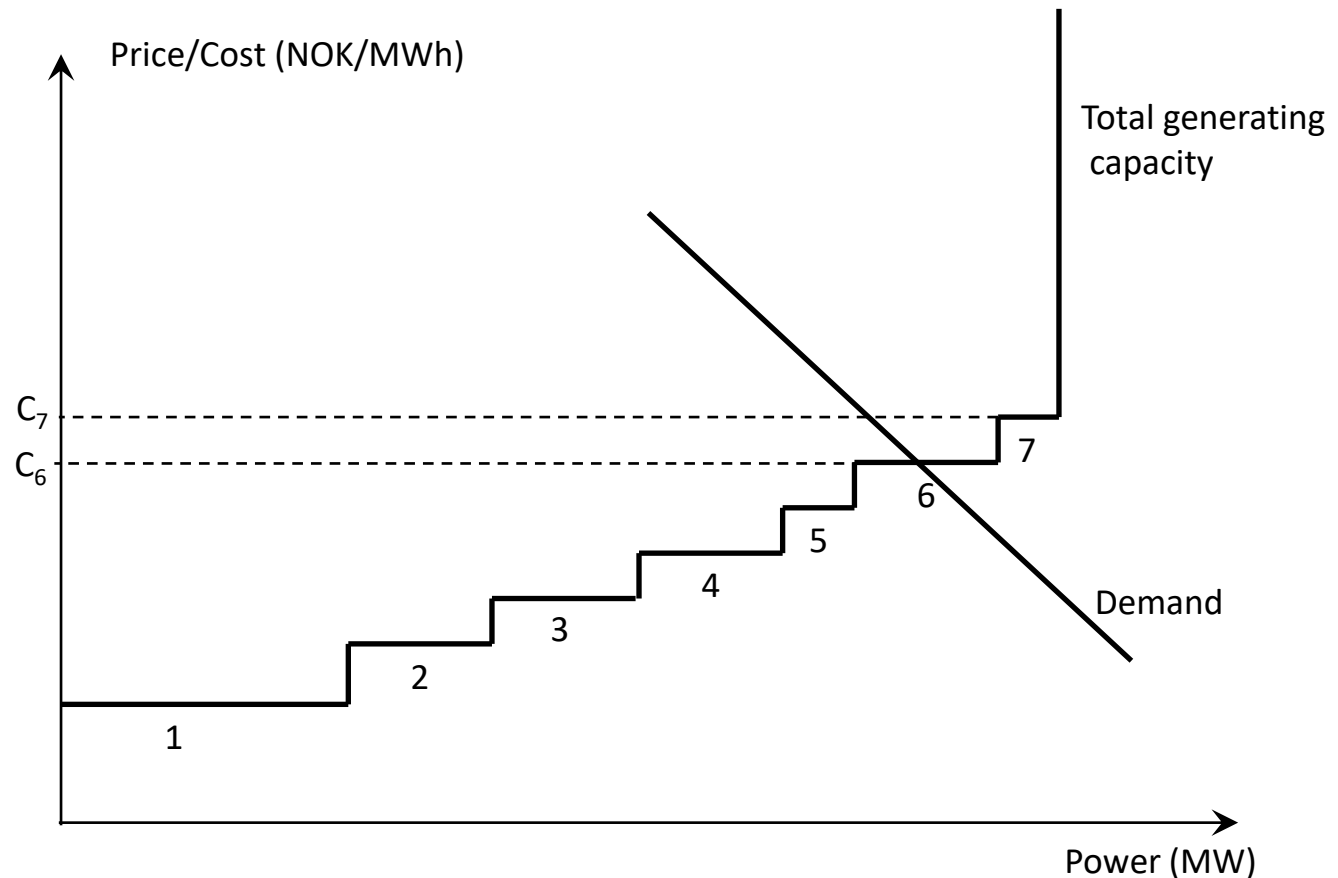
$$\text{Revenue}(\text{“Company A”}) = 10 \times 100 + 100 \times 200 = 21000\text{NOK}$$

- “Company A” gets a capacity reservation of 10MW for **upward** regulation and 20MW for **downward** regulation, it can then only offer between 20 and 90MWh on the day-ahead electricity market
- Say capacity is reserved at 100NOK/MW and all energy sold at $\lambda^S = 200\text{NOK/MWh}$, the resulting revenue is

$$\text{Revenue}(\text{“Company A”}) = 30 \times 100 + 90 \times 200 = 21000\text{NOK}$$

Pricing of reserves based on opportunity cost

No requirement for reserve capacity



Balance between spot market and reserve option market

Spot Market:

Net income	=	Capacity • (spot price – marginal cost)
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Reserve markets:

Net income	=	Capacity • Reserve Option price
	–	Operating cost of keeping reserves available for generation
	+	Expected revenue from possible activation (in the Balancing Market)

- Assumptions:
 - no start-up costs
 - disregard availability cost
 - disregard expected revenue from activation

Spot Market:

Net income	=	Capacity • (spot price – marginal cost)
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Reserve markets:

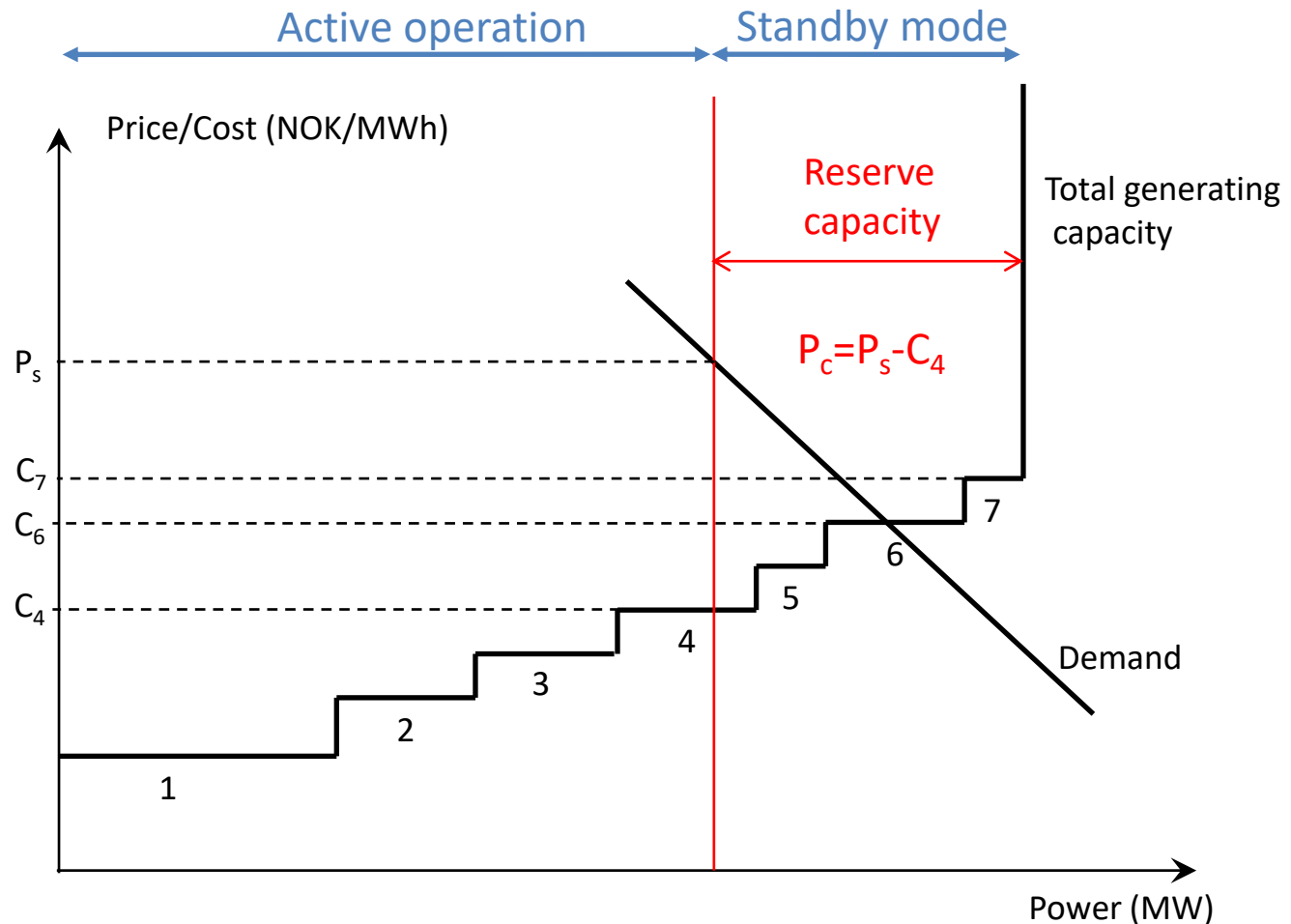
Net income	=	Capacity • Reserve Option price
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- For marginal unit, income from two markets must be equal:

Reserve Option price	=	spot prices – marginal cost
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- Assumptions are very simplified, but still the approach gives a reasonable indication of the cost of providing reserves for hydro units

Pricing of reserves based on opportunity cost-economically optimal solution



Difference between European and the US reserve market

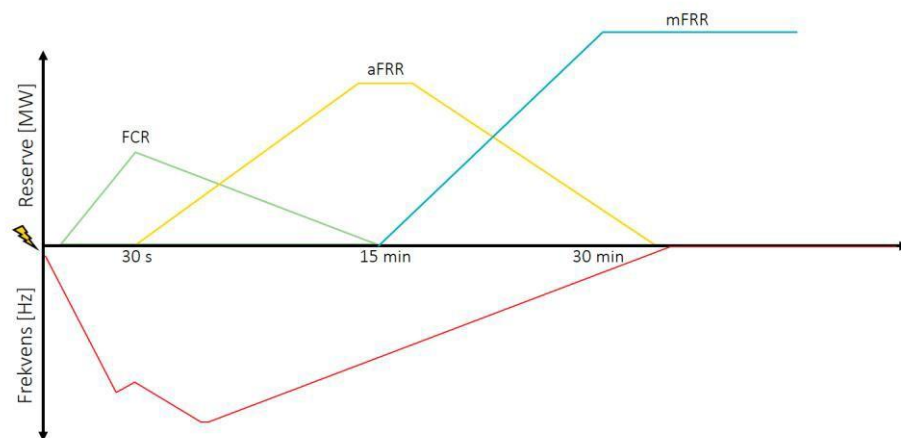
In **European** electricity markets, the reserve and day-ahead energy market are cleared separately (usually, reserve capacity market before the day-ahead energy market):



In the **U.S.** electricity markets, the reserve and day-ahead energy markets are cleared jointly:

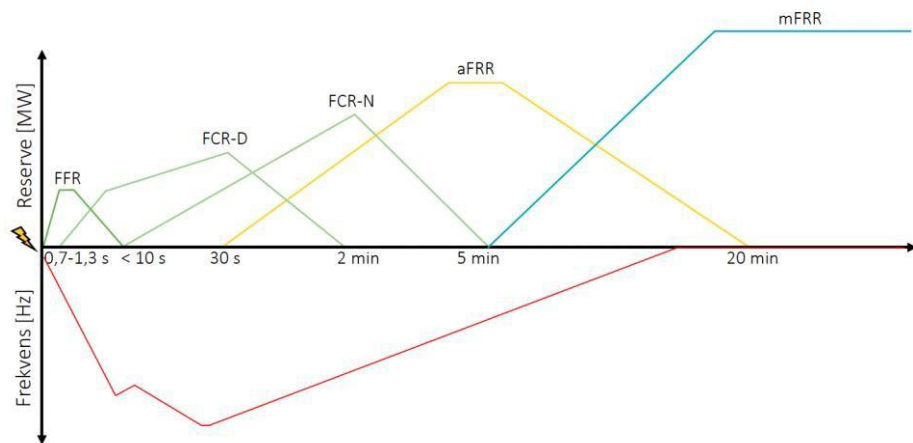


Frequency-based ancillary services in RG Continental Europe (RGCE) and the Nordic



Ancillary services in EU (as part of the continental area)

- FCR
- aFRR
- mFRR



Ancillary services in Nordic (as part of the Nordic area)

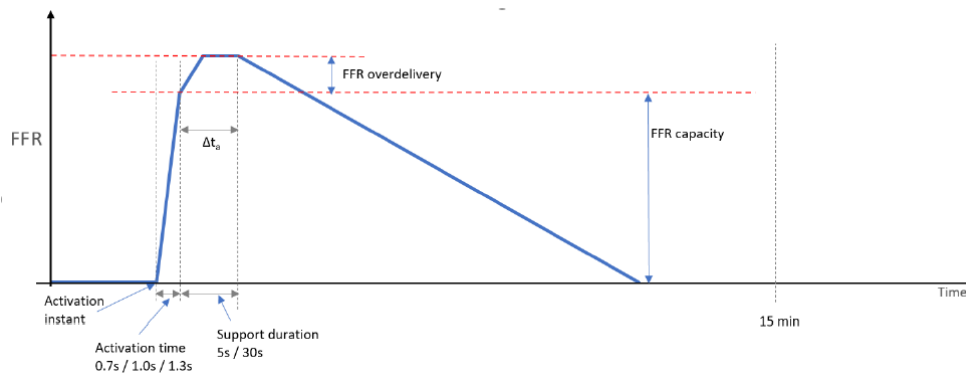
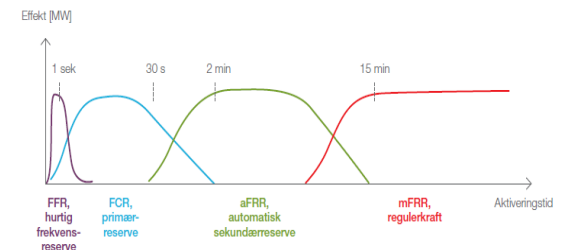
- FFR (fast frequency reserve)
- FCR-D (D stands for disturbance)
- FCR-N (N stands for normal)
- aFRR
- mFRR

Source: Energinet (*Gennemgang af Nuværende Systemydelse Markeder*)

Fast frequency reserves (Not within the scope of this lecture)

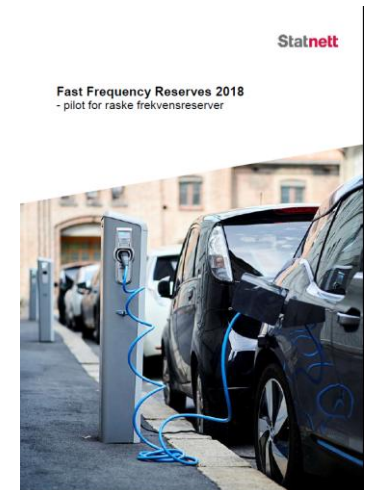
- **Fast frequency reserves (FFR):** The activation happens automatically and very quickly to prevent frequency drops below 49.0 Hz in the event of major faults in situations with low rotational energy/stability in the power system
- Alternatives for activation of FFR

Alternative	Activation level [HZ]	Maximum full activation time [s]
A	49.7	1.3
B	49.6	1.0
C	49.5	0.7



Statnett, as the Norwegian system operator, has committed to procuring 150 MW of FFR in 2021.

<https://www.statnett.no/globalassets/for-aktorer-i-kraftsystemet/utvikling-av-kraftsystemet/smup/systemdrifts-og-markedsutviklingsplan-2022-2030.pdf>
<https://www.statnett.no/for-aktorer-i-kraftbransjen/systemansvaret/kraftmarkedet/reservemarkeder/ffr/>



Nordic various categories of reserve products: response time and duration



1. Fast Frequency Reserves (FFR)

- Brakes the frequency drop
- Response Time: 0.7-1.3 seconds; Duration: 5-30 seconds
- Market for capacity and compensation for activation



2. Frequency Containment Reserves (FCR-N* and FCR-D**)

- Stabilizes frequency to maintain it at a new level
- Requires a dynamic response to counteract frequency variations; 30-120 seconds response; minimum duration 15 minutes
- Market for capacity and compensation for activated volume



3. Automatic Frequency Restoration Reserves (aFRR)

- Restores frequency to normal range (49.9-50.1 Hz)
- Full response within 5 minutes; duration for the period(s) the bid applies
- Capacity market and compensation for activated volume soon market



4. Manual Frequency Restoration Reserves (mFRR)

- Restores and maintains balance until a new balance is reached in the energy market
- Full response within 12.5 minutes; duration for the period(s) the bid applies
- Capacity market (mFRR CM) and activation market (mFRR EAM)

- *FCR-N refers to normal reserves (therefore, the conditions for frequency in normal operation apply, which is up to ± 0.1 Hz above or below 50 Hz)
- **FCR-D refers to disturbance reserves (therefore, it applies for conditions outside the normal operation range)

Read more about the conditions for participation, implementation guides, and prequalification documents:

- ✓ [Fast frequency reserves – FFR](#)
- ✓ [Primærreserver – FCR](#)
- ✓ [Sekundærreserver – aFRR](#)
- ✓ [Tertiærreserver – mFRR](#)

Frequency-based ancillary services in RGCE and the Nordic



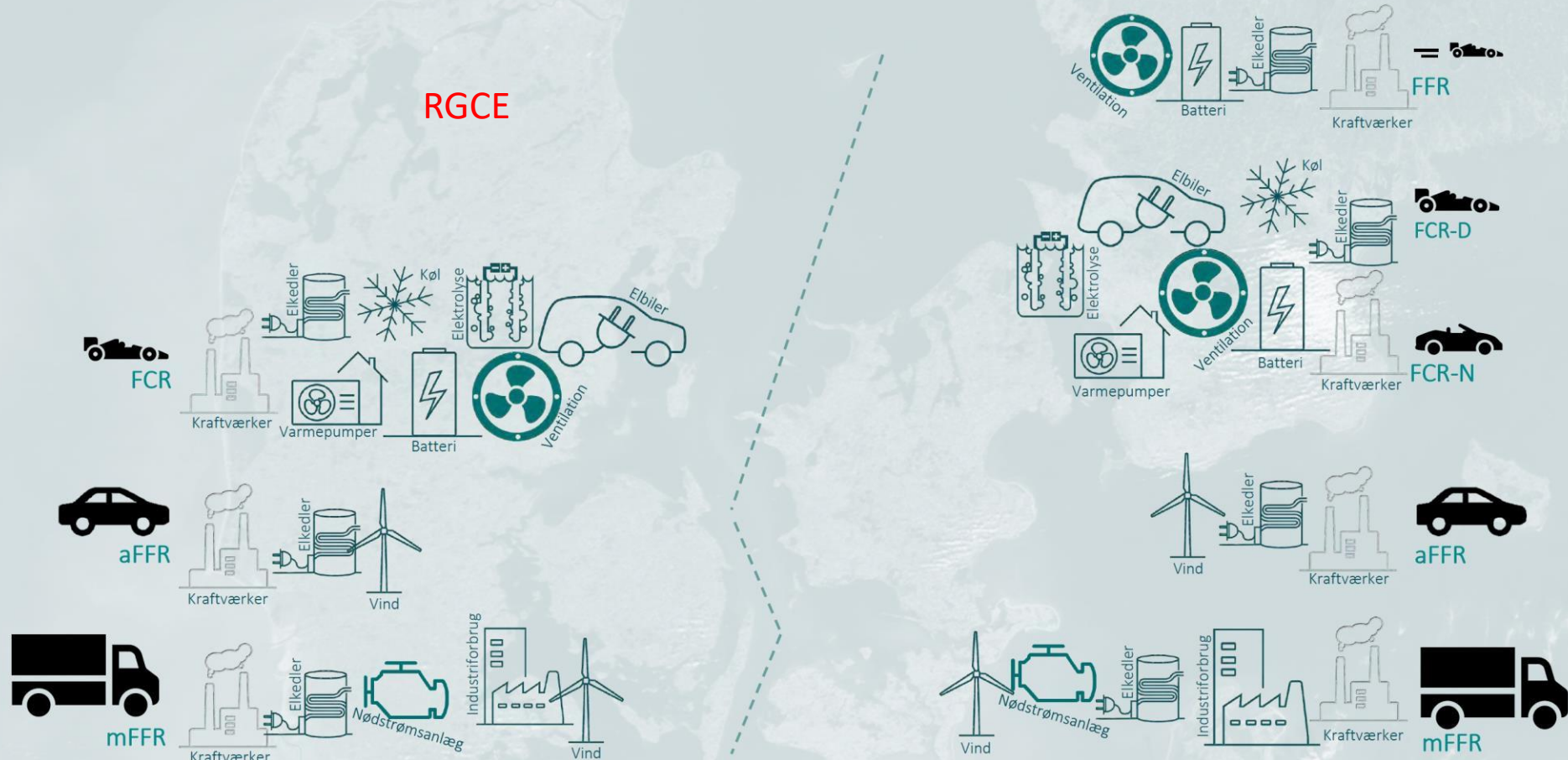
Frequency-based ancillary services in RGCE and the Nordic

SYSTEMYDELSER: TEKNOLOGIERNE BAG

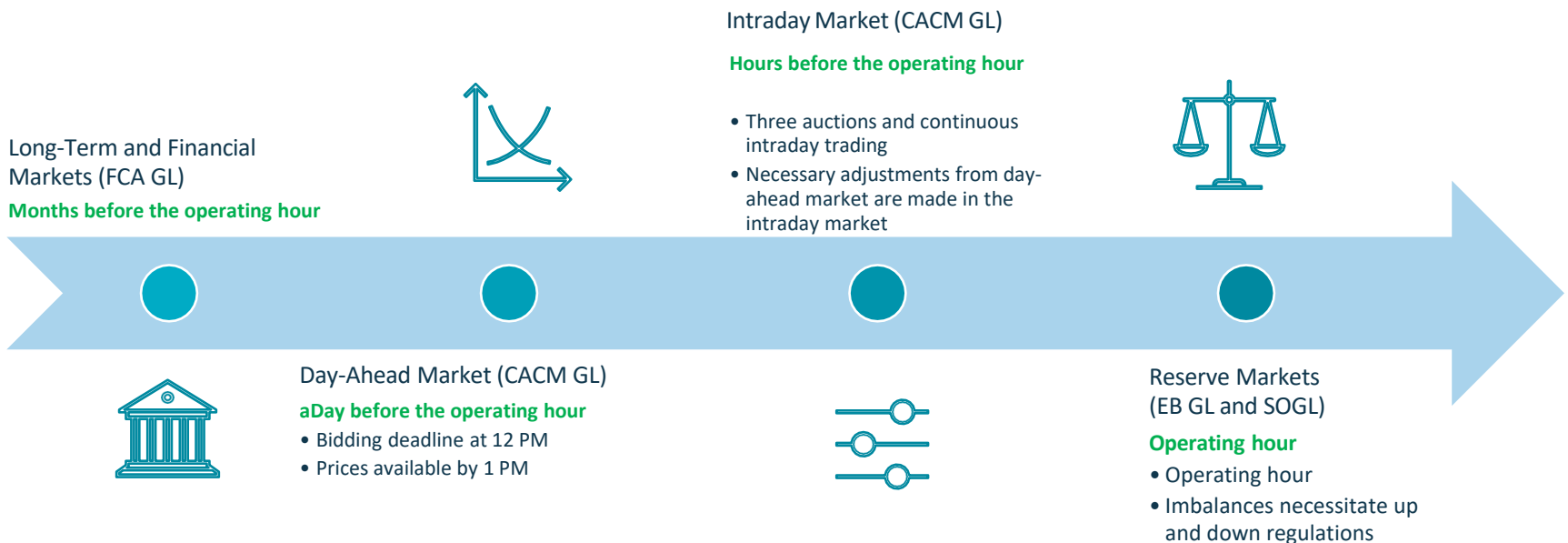
ENERGINET

Nordic

RGCE



Operation of the Power System Through a Sequence of Markets (Nordic)



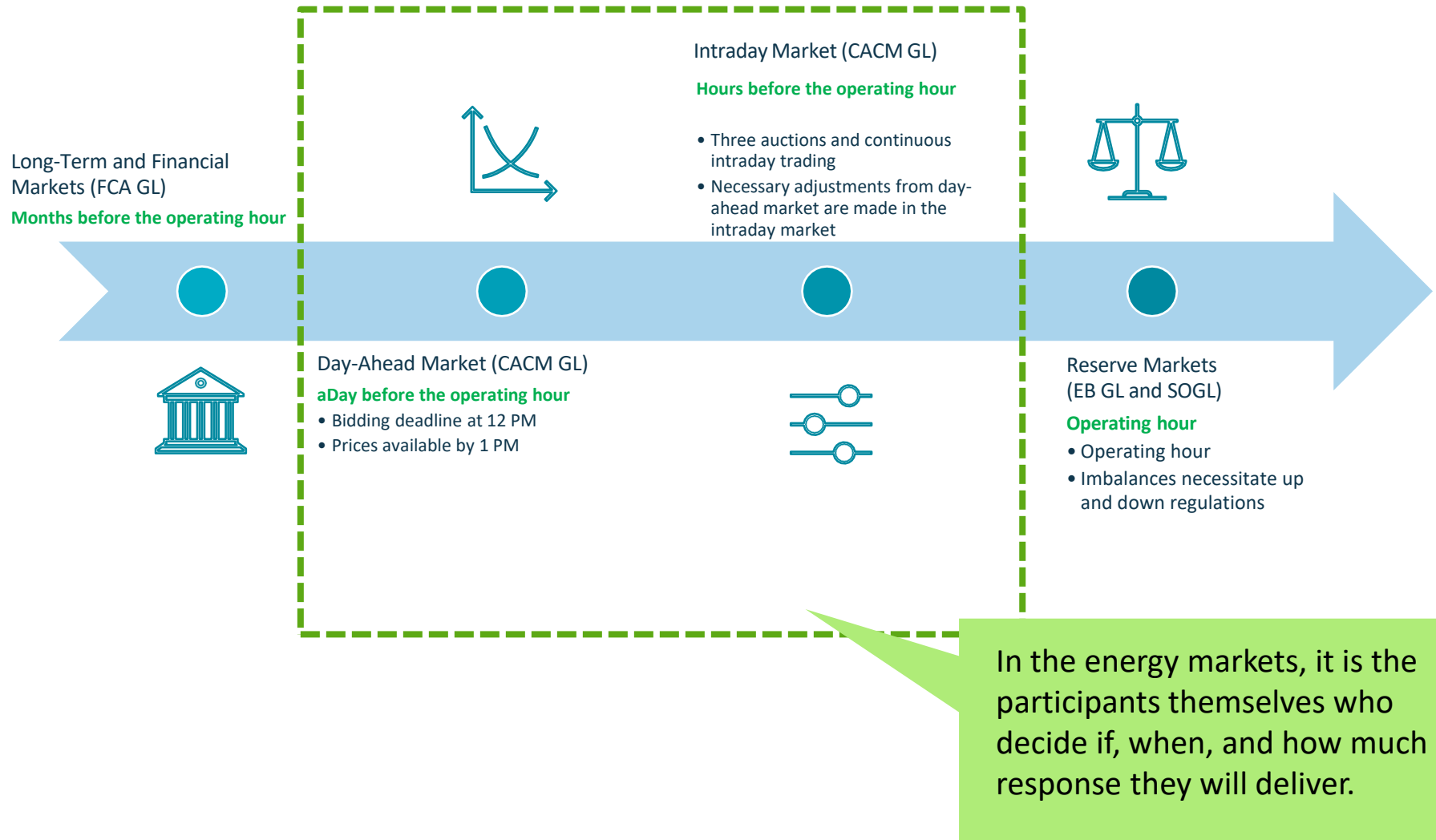
FCA GL: Financial Markets Guideline

CACM GL: Capacity Allocation and Congestion Management Guideline

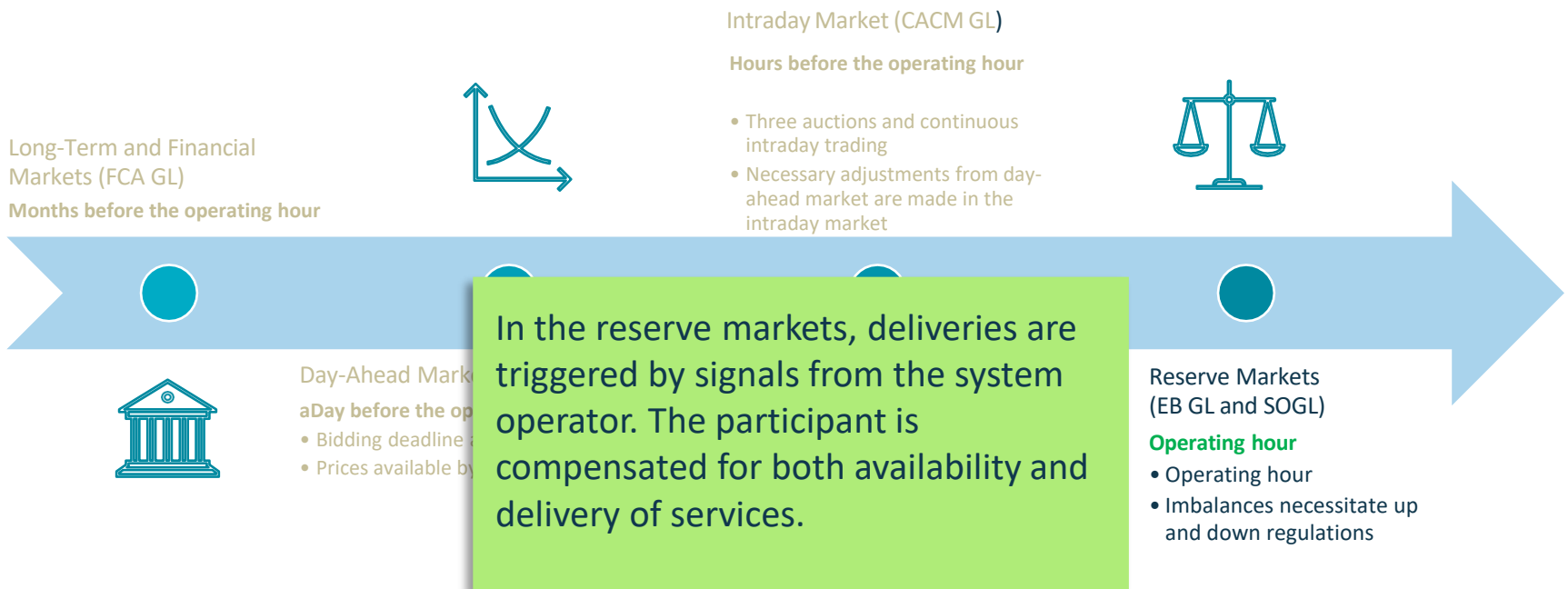
EB GL: Electricity Balancing Guideline

SOGL: System Operation Guideline

Operation of the Power System Through a Sequence of Markets (Nordic)



Operation of the Power System Through a Sequence of Markets (Nordic)

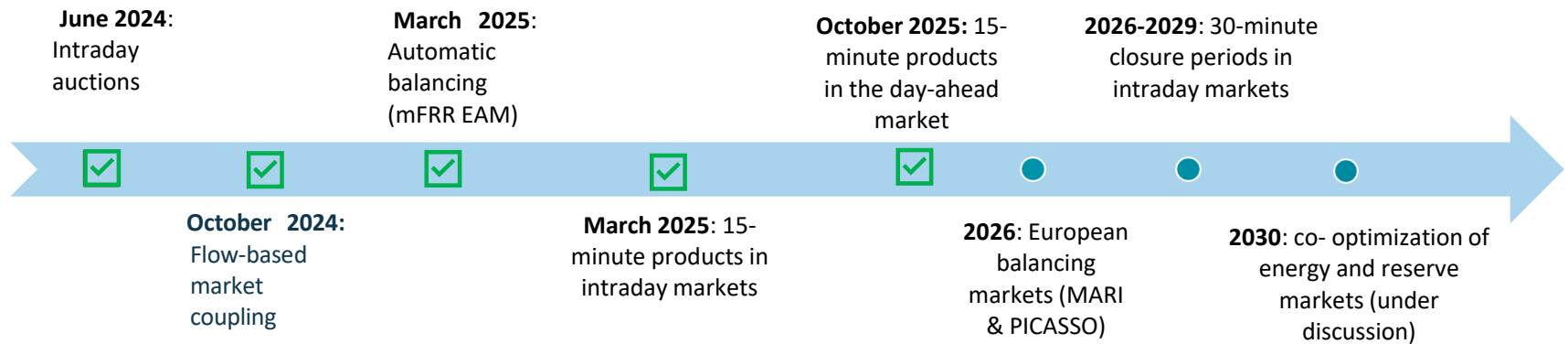


Reserve market

- Reserve Capacity Markets Ensure Reserves
- Bidding deadlines and clearing in reserve capacity markets occur before the deadline for the day-ahead market
- Trading capacity is reserved from the day-ahead market for the

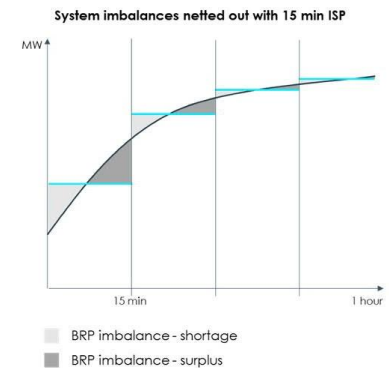
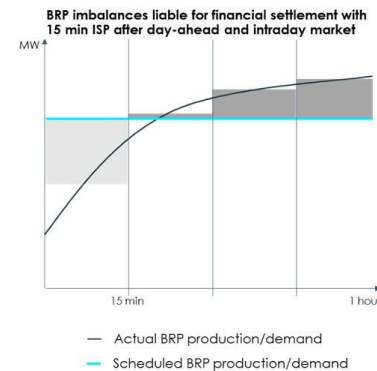
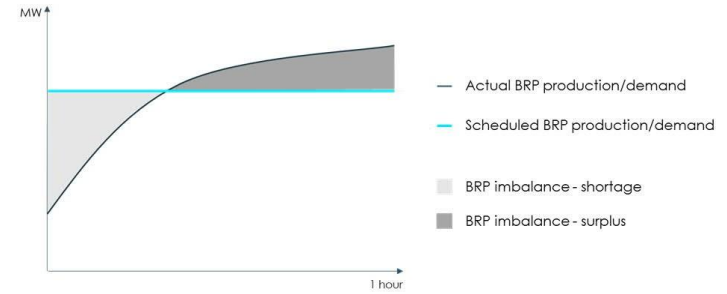
- Reserves are activated in the activation market or through automatic response

The Development in Power Reserve Markets is Rapid!



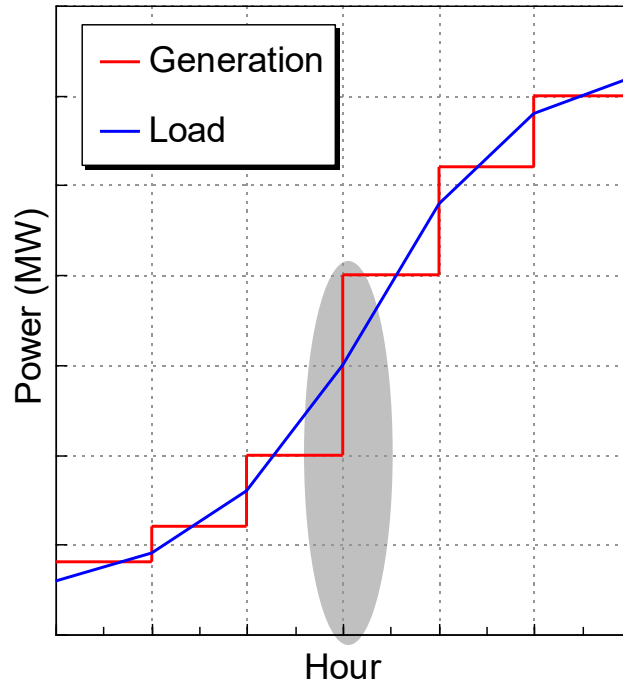
Why 15-minute imbalance settlement period (ISP)

- Market model adapts to the changing power system
 - *Green transition* – increasing amount of intermittent energy
- Incentivising maintaining the balance in more *Real-time* for a more cost-efficient power system
- Increased possibilities for ancillary services
- Electricity market harmonization in Europe
- European legislation (EBGL- Electricity Balancing Guideline)



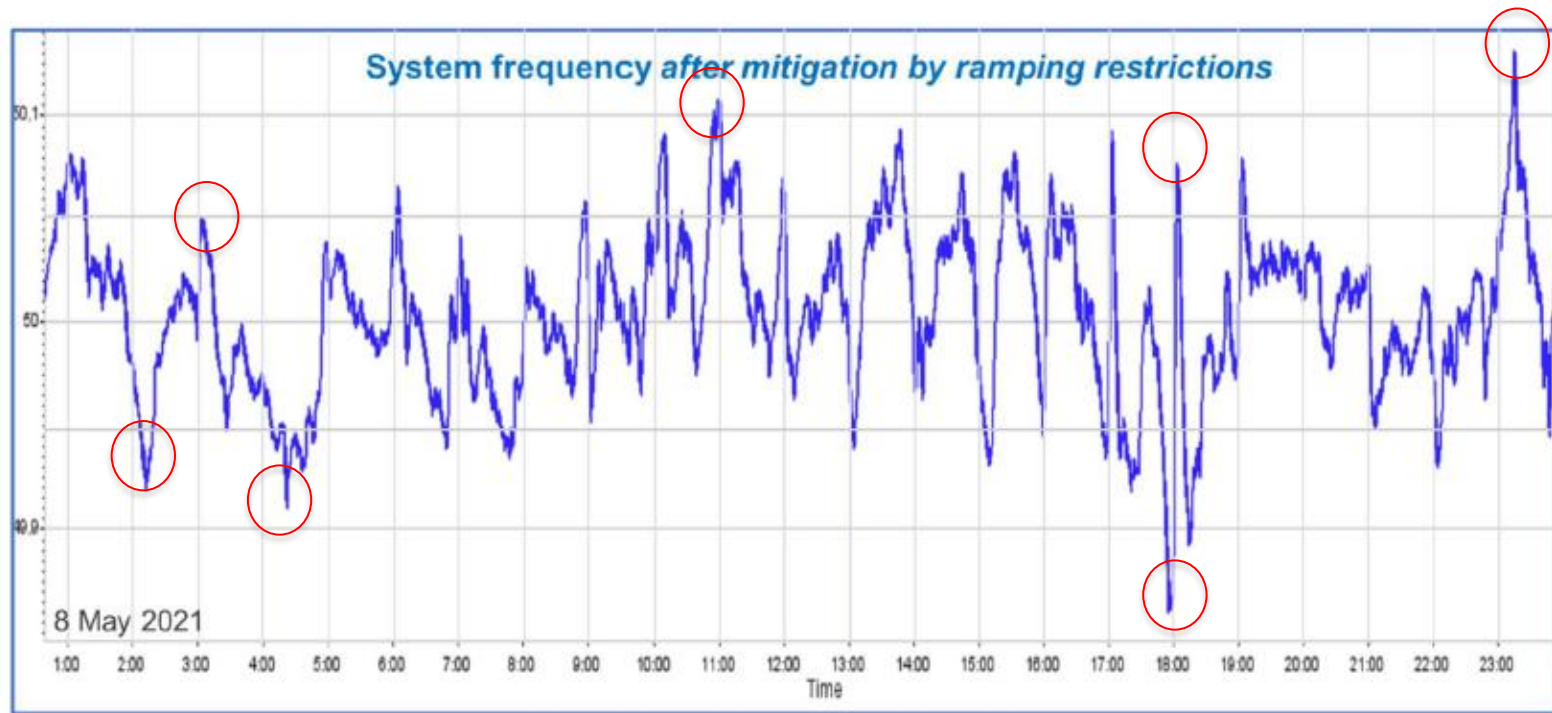
Source: NBM

Structural imbalances



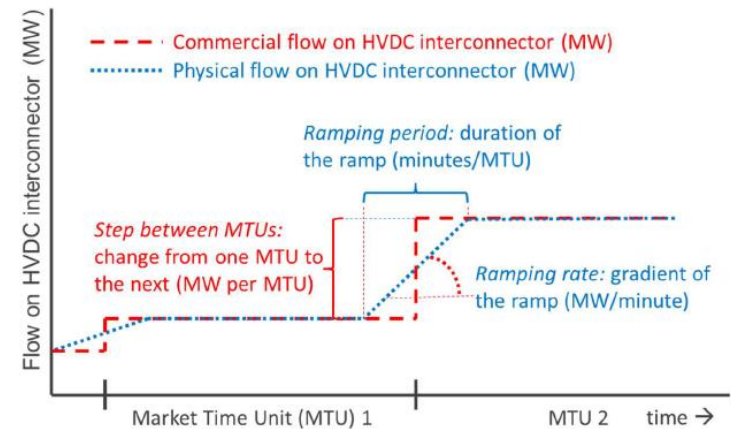
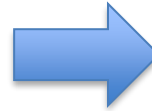
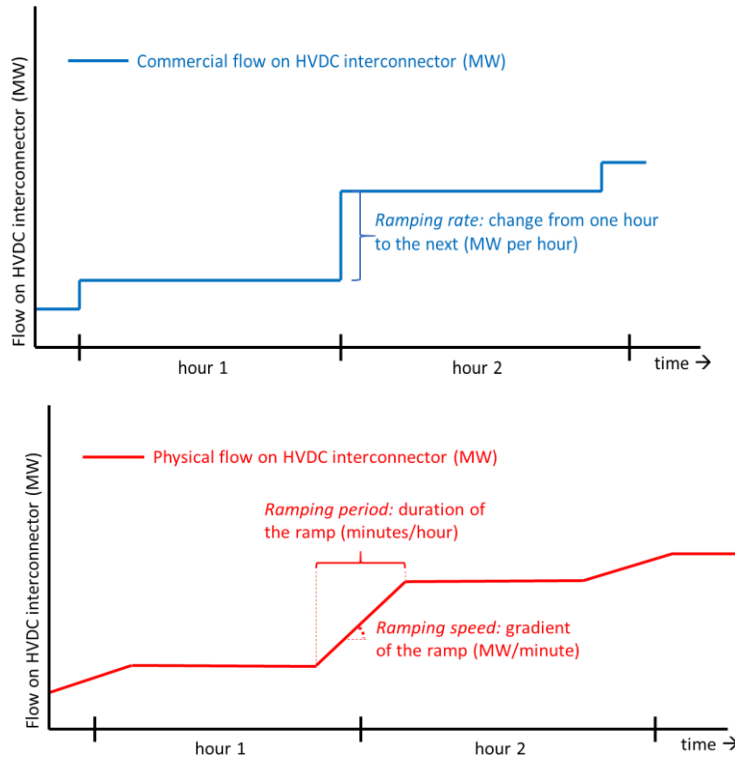
- Generation units are normally scheduled to adjust or start up every new hour according to the spot market clearing (MWh/h)
- The load changes continuously.
 - Consumption increases in the morning and decreases in the evening.
 - The interconnectors often ramp from import to export in the morning, and from export to import in the evening.

System Frequency in the Nordic Synchronous area



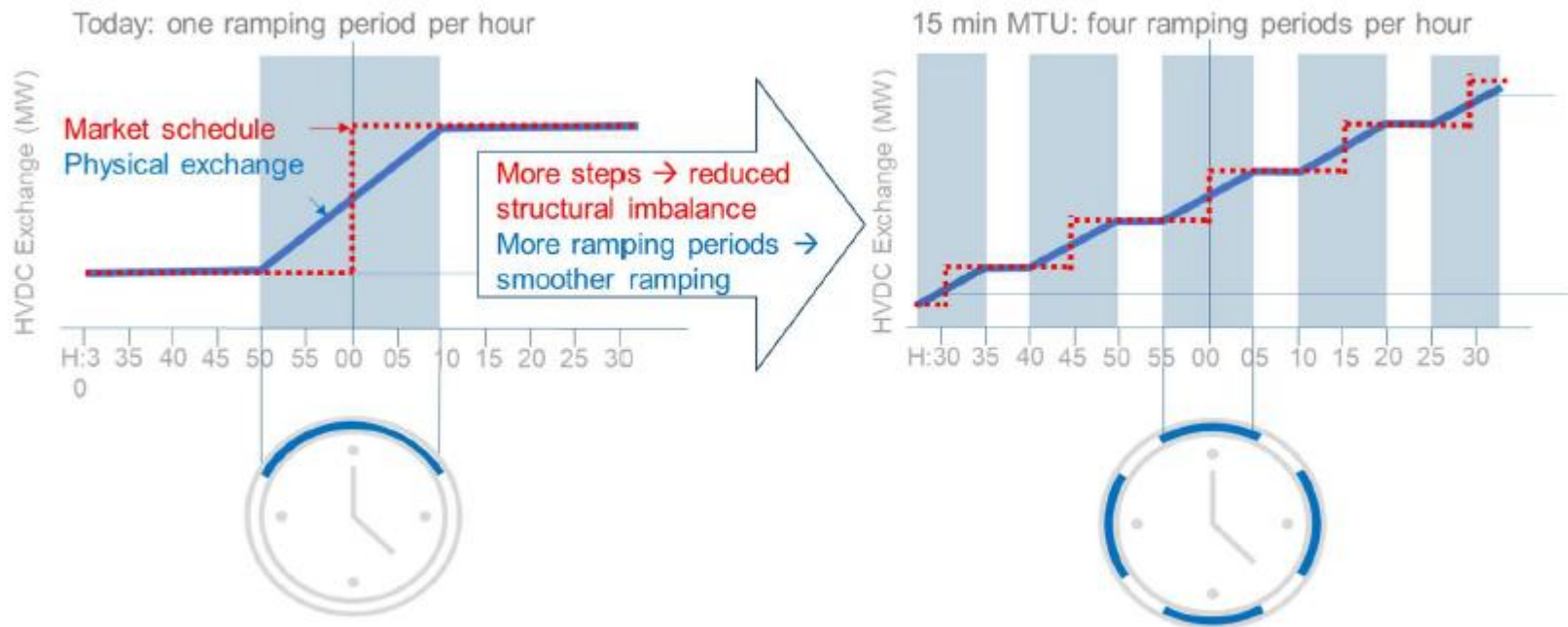
- Sharp peaks around the times that the market schedules change, which are currently hour shifts.
- The peaks are called deterministic frequency deviations (DFDs) or Structural imbalance
- They are caused by the momentary structural imbalances that result from differences in ramping between electricity demand, production and HVDC exchange to other synchronous areas.

Ramping on HVDC cables



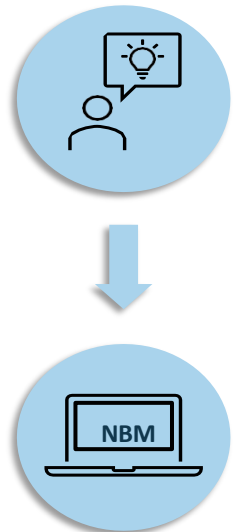
- Maximal ramping speed (rate) is 30 MW/min
- The change of load on the cable (ramping) occurs continually within the twenty-minute period around the hourly shift.

Smoother ramping with 15 minutes MTUs in the cross border intraday and day-ahead markets

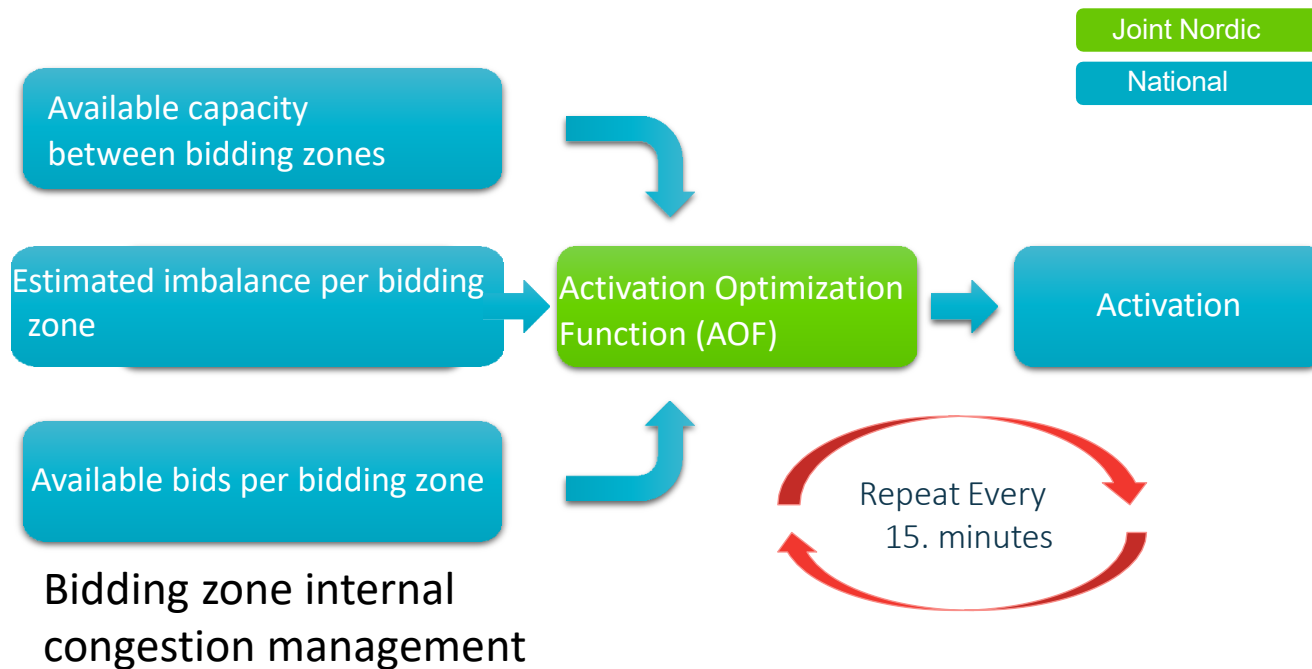


Transitioning to Automated Balancing

- **Current Balancing Process Challenges**
- No longer straightforward due to:
 - Greater fluctuations in renewable energy sources causing rapid and significant changes in balance both within Norway and across borders.
 - Increased frequency of resource regulation in the market.
 - Numerous smaller reserves distributed geographically.
 - Introduction of 15-minute resolution in energy markets.
- **Shift from Operator to Computerized Systems**
 - Estimate needs, automatic bid selection and activation, standardized response
 - Reduced time for evaluations and operator interventions
 - More transparent system operations, but less flexible.
 - Primarily a cost-reducing measure.
- **Implications**
 - This represents a major shift affecting the entire energy sector.

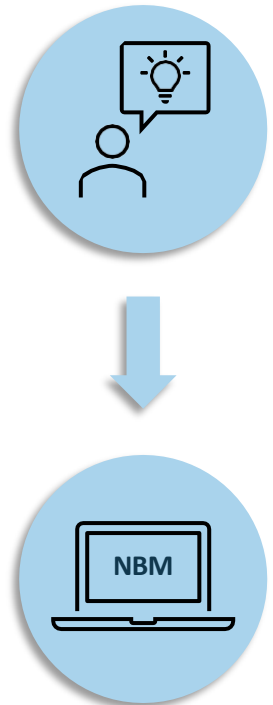


The automated balancing process



Automated balancing influences many processes

- **Bottleneck Management to be Automated**
 - We utilize the same bids for resolving bottlenecks as for balancing.
 - Activate system adjustments: Are there bottlenecks in the system that need control?
 - Filter bids: Are there bids that could create bottlenecks if activated?
 - Bottleneck management must be completed between the bid submission deadline (T-0.45) and the deadline to send bids to the bid selector (T-0.12).
- **New Dimensioning Method for reserves**
 - Increased reserve requirements, reserve requirements per area.
 - aFRR is managed on a Nordic level, mFRR nationally.
- **Future Activation Markets are European**
- MARI is similar to Nordic mFRR Energy activation market (EAM).
- PICASSO entirely renews the reserve market: aFRR EAM per price area.
- Exchange opportunities across all of Europe via HVDC.

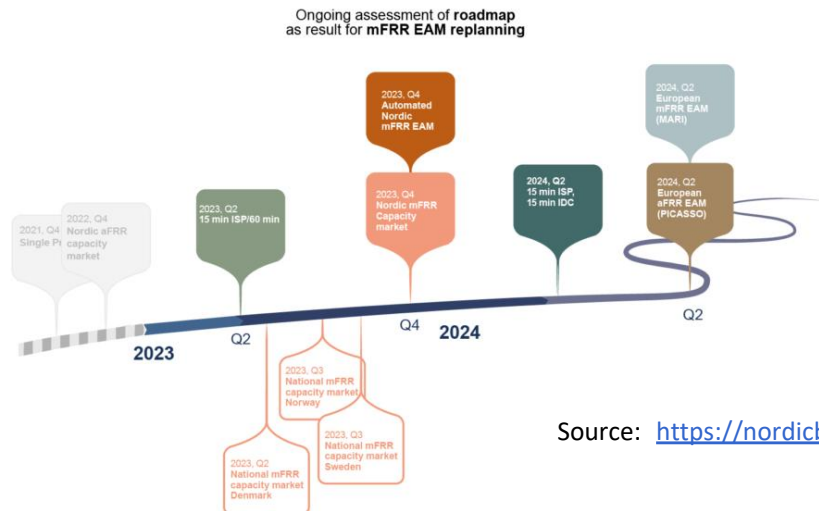


Upcoming pan-European balancing platforms

- **PICASSO**: An exchange platform for **aFRR** between European countries (https://www.entsoe.eu/network_codes/eb/picasso/)
- **MARI**: An exchange platform for **mFRR** between European countries (https://www.entsoe.eu/network_codes/eb/mari/)

The expected go-live date for these European balancing platforms is in Q2, 2026.

Statnett has a close operation and market cooperation with the other Nordic TSO's and they have decided to implement a common project called *Nordic Balancing Model*, **NBM** (<https://nordicbalancingmodel.net/>) to eventually join PICASSO and MARI.



Source: <https://nordicbalancingmodel.net/roadmap-and-projects/>

From frequency-based to ACE-based

- Today: We activate for the net imbalance in the so-called mutually regulated or uncongested area. We do not know the imbalance of an individual bidding zone, and are only able to activate on the net imbalance. We mainly use mFRR to balance (together with the "frequency-band") and only have one price-setting product.
- Future: With ACE-based balancing, we can identify the **imbalance of each bidding zone** and **forecast the mFRR demand** of each bidding zone. It is possible to optimise how to meet this demand (by netting or activation) taking into account **available transmission capacity (ATC)** and bid prices. It is not always economical to net. We will use and have several price-setting products with MARI and Picasso. ACE-based balancing is a prerequisite to join MARI and Picasso.

Before, March 2025

- Manual process
- 60 min (24 gate closures/day)
- Time to agree and actions on phone
- Call IT, when needed

Now

- 15 min (96 gate closures/day)
- Small bid size
- Automated bid selection process
- Electronic ordering
- Situational awareness
- Intervene only when needed
- IT support in control rooms

PICASSO-The Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation

The main targets of the project are:

- Design, implement and operate an aFRR-Platform compliant with the approved versions of European network code regulations.
- Enhancing economic and technical efficiency within the limits of system security.
- Integrating the European aFRR markets while respecting the TSO-TSO model.

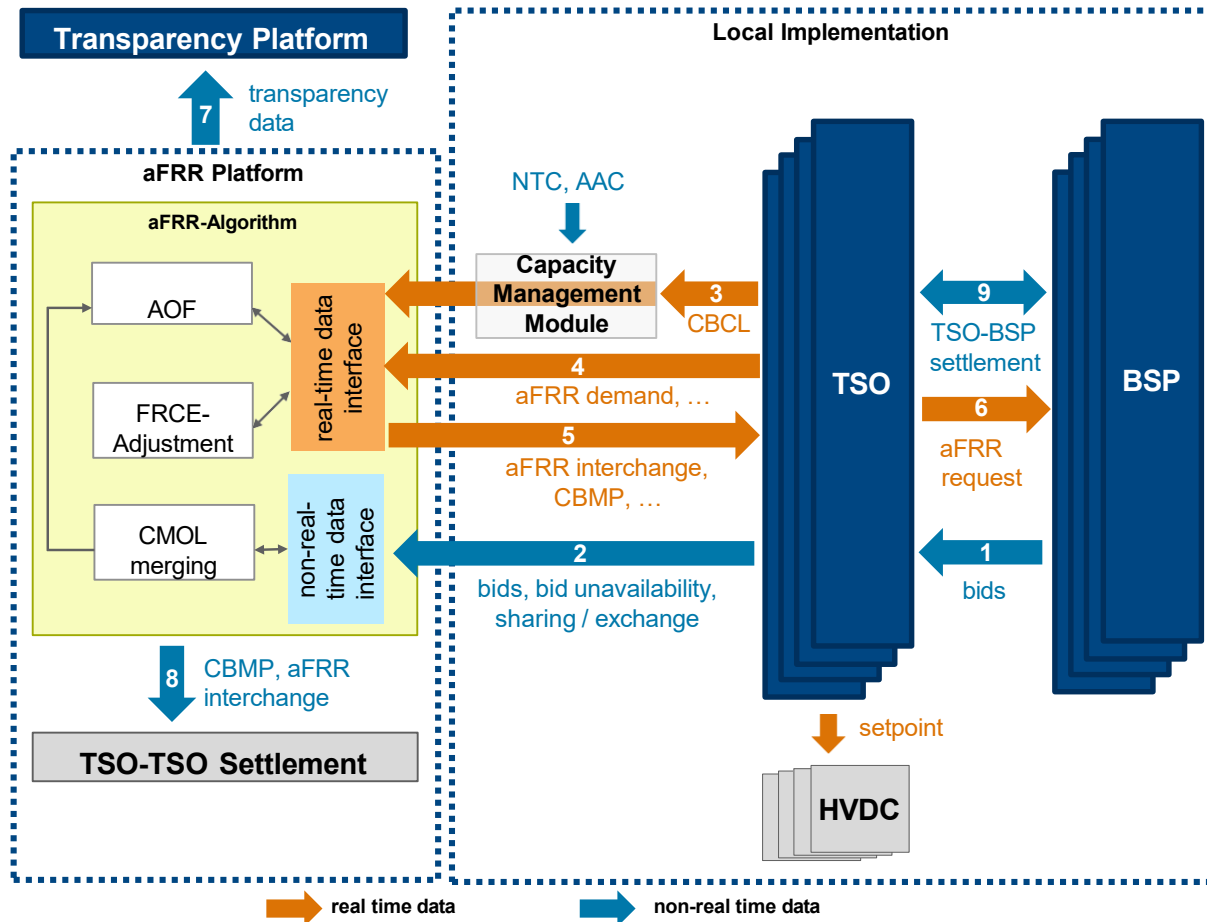
30 TSOs + ENTSO-E (Observer)		
PICASSO Members (26 TSOs)		PICASSO Observers (4 TSOs + ENTSO-E)
Austria 	Hungary 	Latvia 
Belgium 	Italy 	Lithuania 
Croatia 	The Netherlands 	Estonia 
Czech Republic 	Norway 	North Macedonia 
Denmark 	Poland 	ENTSO-E 
Finland 	Portugal 	
France 	Romania 	
Germany 	Slovak Republic 	
	Slovenia 	
Sweden 	Spain 	
Bulgaria 	Greece 	
Switzerland 	Luxembourg 	



PICASSO Platform design



PICASSO Platform design - PICASSO High Level Design



1. TSO receives bids from BSPs in their LFC area
2. TSO forwards standard aFRR balancing bids to platform
3. TSOs communicate Cross-Border Capacity Limits to Platform
4. TSOs communicate aFRR demands to platform
5. Communication of clearing results to TSO
6. Communication of aFRR request from each LFC to BSP
7. Data Publication
8. TSO-TSO settlement process and invoicing
9. TSO-BSP settlement process and invoicing

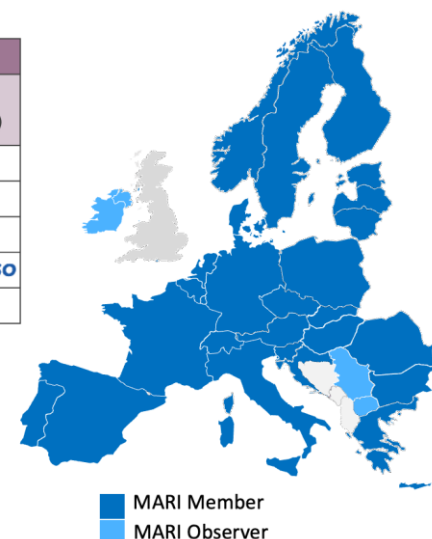
AOF Activation Optimisation Function
 FRCE Frequency Restoration Control Error
 CMOL Common Merit Order List
 CBMP Cross-Border Marginal Price

NTC Net Transfer Capacity
 AAC Already Allocated Capacities
 CBCL Cross-Border Capacity Limits
 HVDC High Voltage Direct Current

MARI - Manually Activated Reserve Initiative

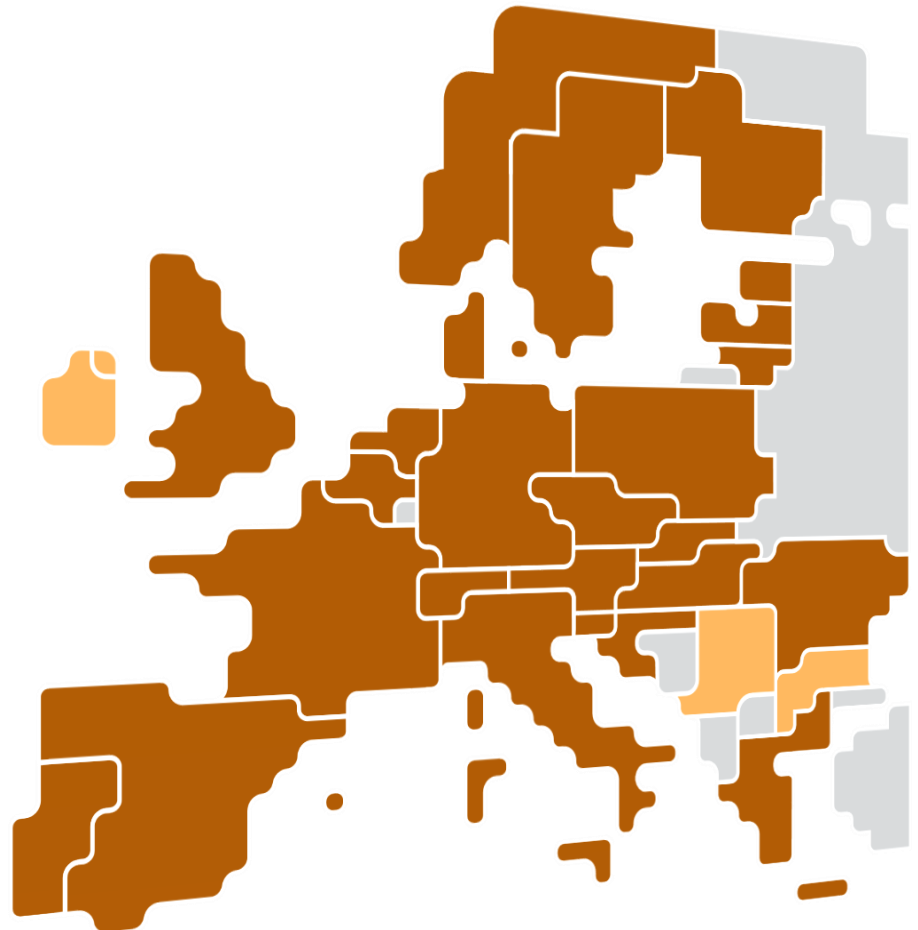
- Voluntary initiative with 33 TSOs to design, implement and operate a platform for manual frequency restoration reserves (mFRR)
- Aims at designing and implementing an mFRR Platform which should become the European solution
- The platform has been brought successfully into operation on 5 October 2022 with accession of 5 TSOs (Czech Transmission System Operator (CEPS) and the German TSOs (TenneT DE, 50Hertz, Amprion, and TransnetBW))

33 TSOs + ENTSO-E (Observer)					
MARI Members (29 TSOs)				MARI Observers (4 TSOs + ENTSO-E)	
Austria	APC	Greece	ASST	Spain	RED ELÉCTRICA DE ESPAÑA
Belgium	Elia	Italy	ENEL	Sweden	SVERIGES KRAFTFÖRSÄLLING
Croatia	HOPS	Latvia	AST	Switzerland	swissgrid
Czech Republic	ČEPS	Lithuania	LEI	Northern Ireland	SCHNI
Denmark	ENERGINET	Norway	Statnett	Luxembourg	CREOS
Estonia	elering	Netherlands	tennet	Bulgaria	БЕЛ
Finland	FINGRID	Portugal	REN	ENTSO-E	entsoe
France	enx	Poland	PSE		
Hungary	MOL	Romania	TRASE		
Germany	amprion 50hertz tennet transnetbw	Slovak Republic	SPP		
		Slovenia	ELES		



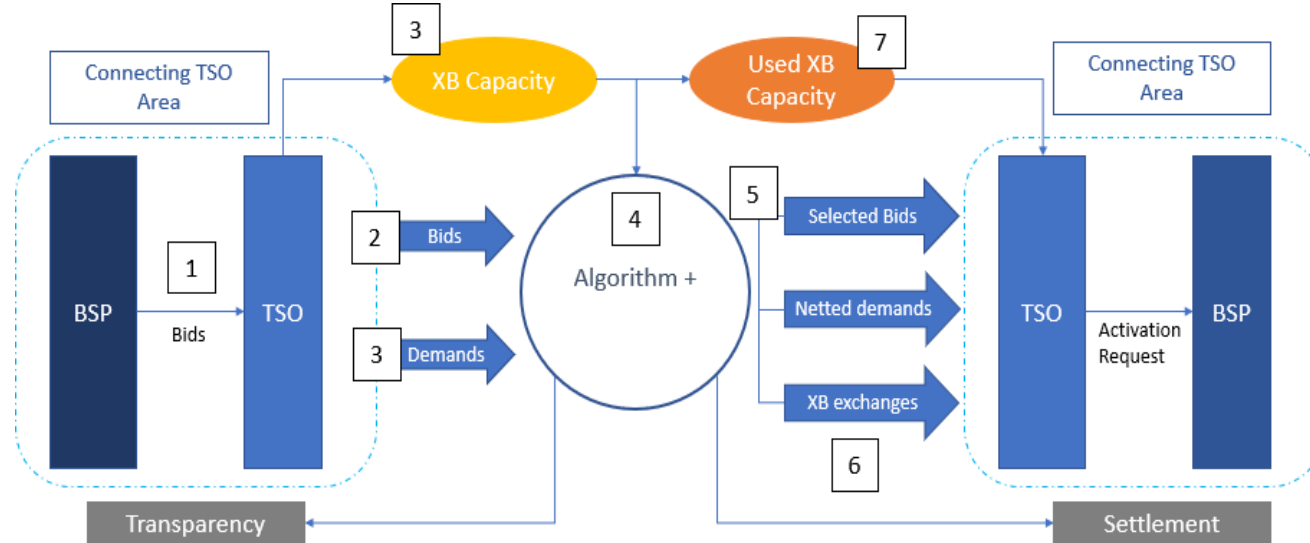


mFRR Process



The mFRR process - General Process of mFRR Activation

The MARI platform consists in (i) a data management platform (Libra Data Management), handling communications and data exchanges with TSOs and ENTSO-E, pre-processing, postprocessing and archiving, and (ii) an optimization module (AOF) optimizing the activations and computing Cross Border Marginal Prices (CBMP).

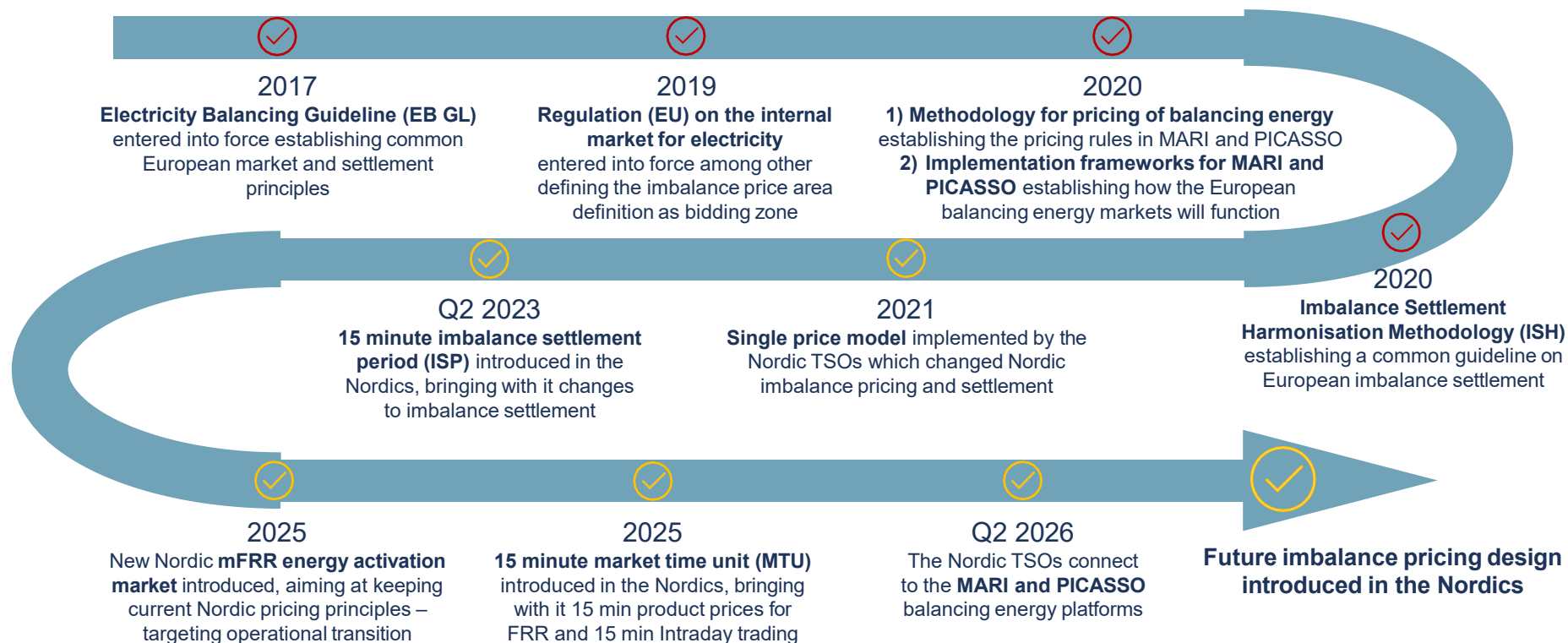


1. TSO receive bids from BSPs in local market balance area
2. Forward of coherent mFRR balancing products to mFRR platform
3. TSOs communicate their balancing demands via the balancing platform and the available XB transmission capacities (ATC) either directly or via CMM*.
4. Optimization of the clearing of balancing demands against BSPs bids
5. Communication of the accepted bids, satisfied demands and prices
6. Calculation of the commercial flow between market balancing areas and settlement of the expenditure and revenues between TSOs
7. The resulting XB schedules and remaining ATC are sent to the TSOs directly and to CMM (if configured).

* CMM is designed as a Single Point of Information when it comes to balancing capacities, which improves transparency among the TSOs. In future, the primary source of ATCs shall be CMM. Direct provisions of ATCs by TSOs to balancing platforms will be a backup option. It is foreseen that TSOs will connect to the balancing platforms first and then to CMM in order to test that the backup of sending ATCs is functioning properly.

<https://nordicbalancingmodel.net/>

The road leading to the needed changes



✓ : Legal methodology or framework implemented

✓ : Market or settlement changes implemented